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Walker

(54) NFC-ENABLED MEDICAL DEVICE SYSTEM AND PROCESSES FOR GENERATING ASSOCIATED INFORMATION LAYER

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(60) Provisional application No. 62/587,879, filed on Nov. 17, 2017, provisional application No. 62/674,565, filed on May 21, 2018, provisional application No. 62/680,116, filed on Jun. 4, 2018.

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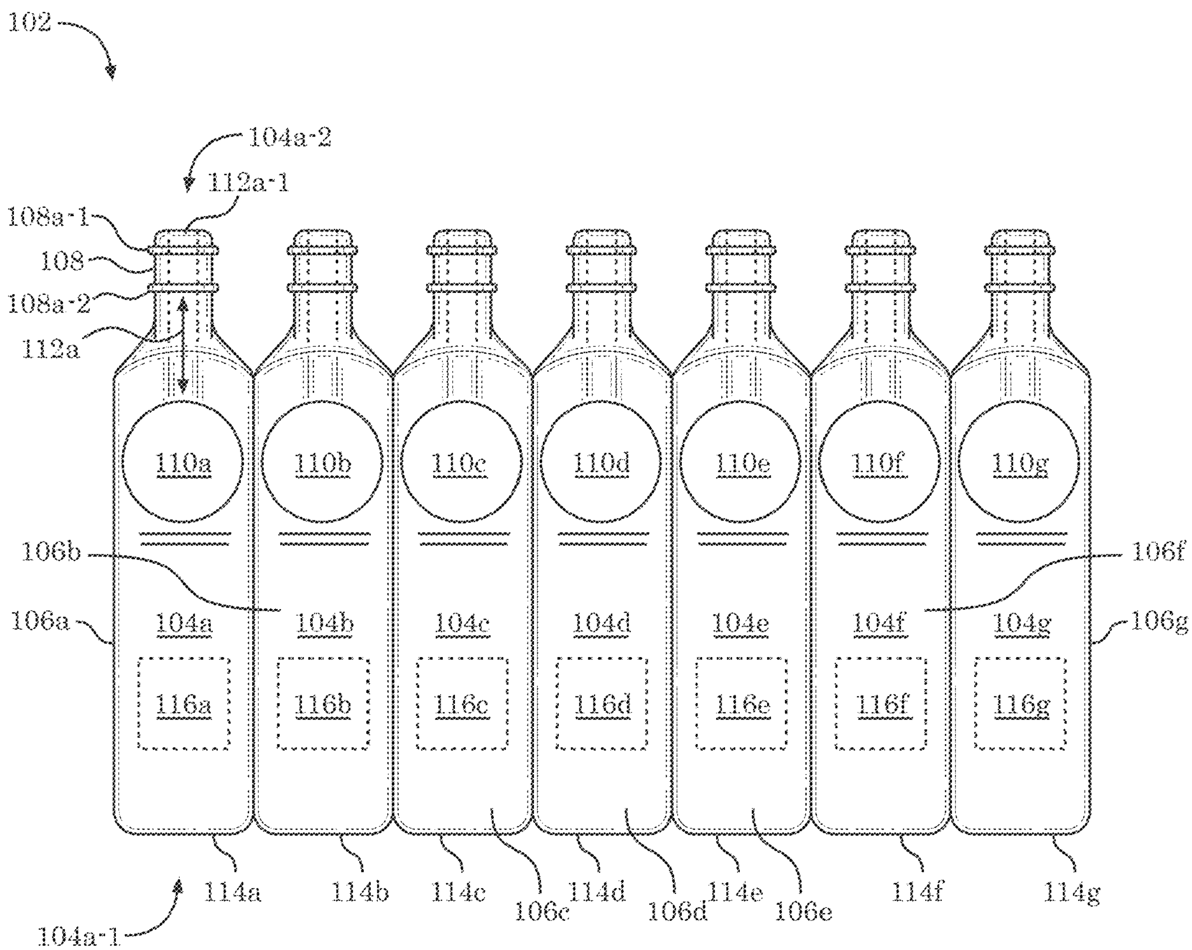
CPC ***A61J 1/067*** (2013.01); ***A61M 5/002*** (2013.01); ***A61M 5/288*** (2013.01); ***A61M 5/3202*** (2013.01); *A61J 2200/72* (2013.01); *A61J 2205/20* (2013.01); *A61J 2205/30* (2013.01); *A61M 2202/30* (2013.01); *A61M 2205/3546* (2013.01); *A61M 2205/52* (2013.01)

(57) **ABSTRACT**

Systems, methods and articles of manufacture provide for an injection management platform that allows the verification and management of injection event transactions involving injectors equipped with NFC or RFID chips, generating and/or reading data via such NFC or RFID chips and authorizing injections based on such data.



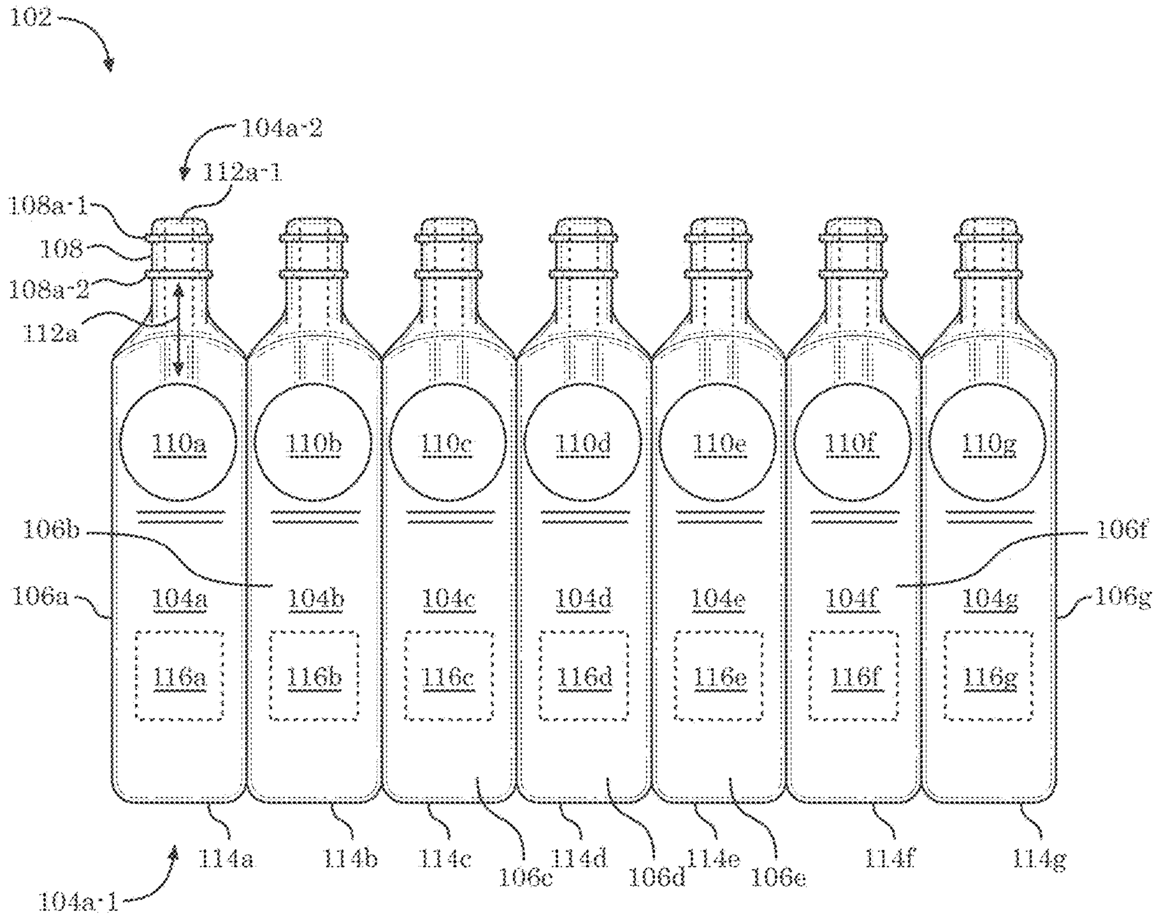

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FIG. 1

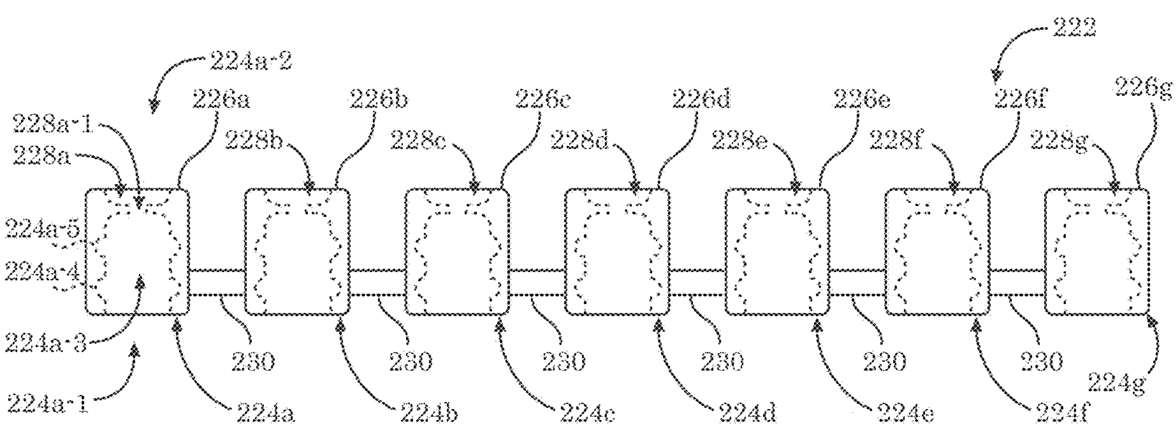
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FIG. 2

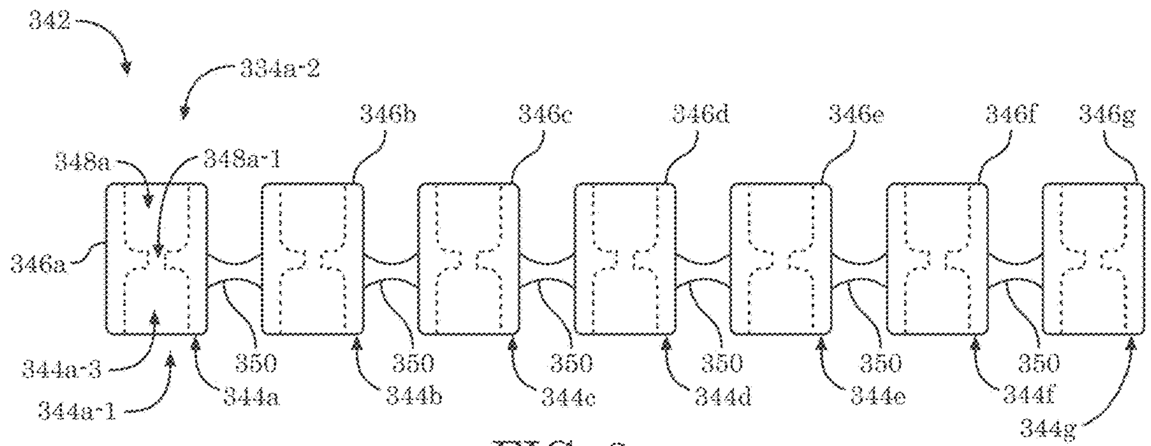

FIG. 3

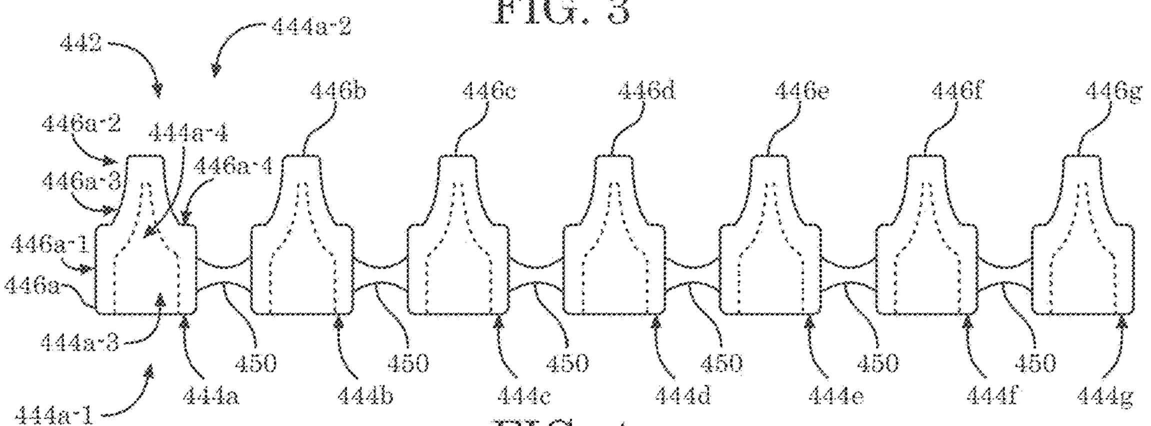

FIG. 4

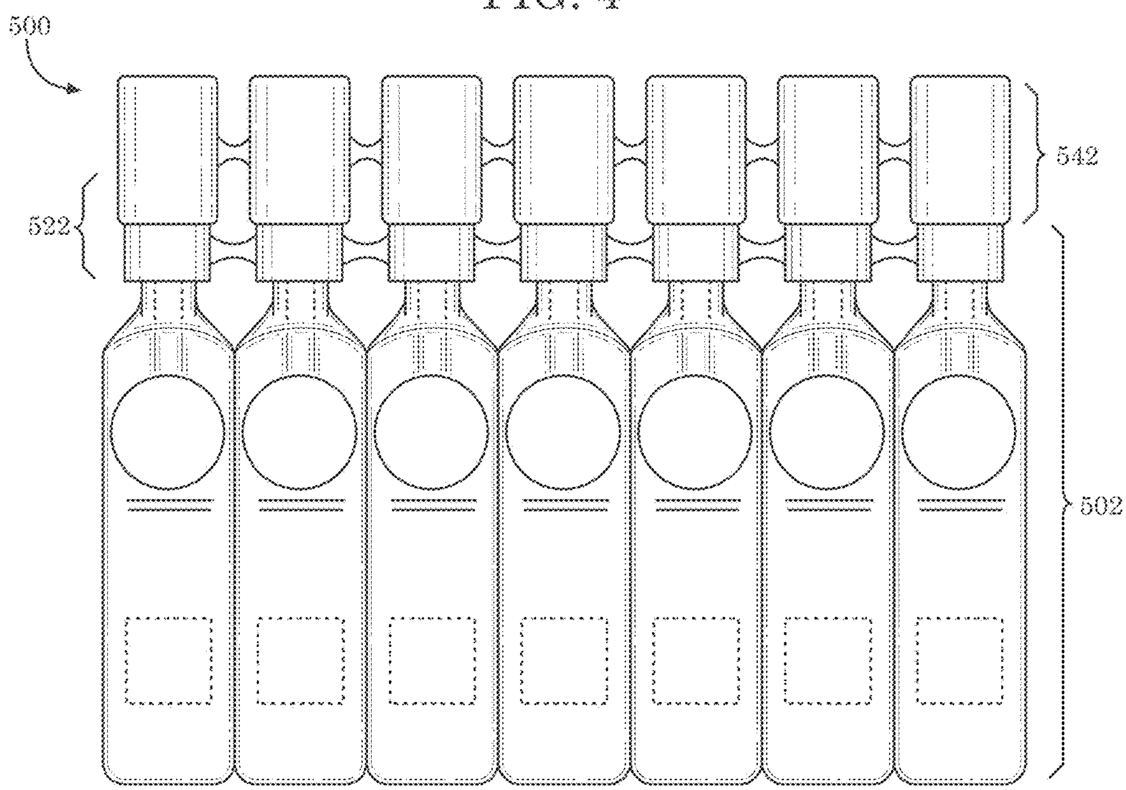

FIG. 5A

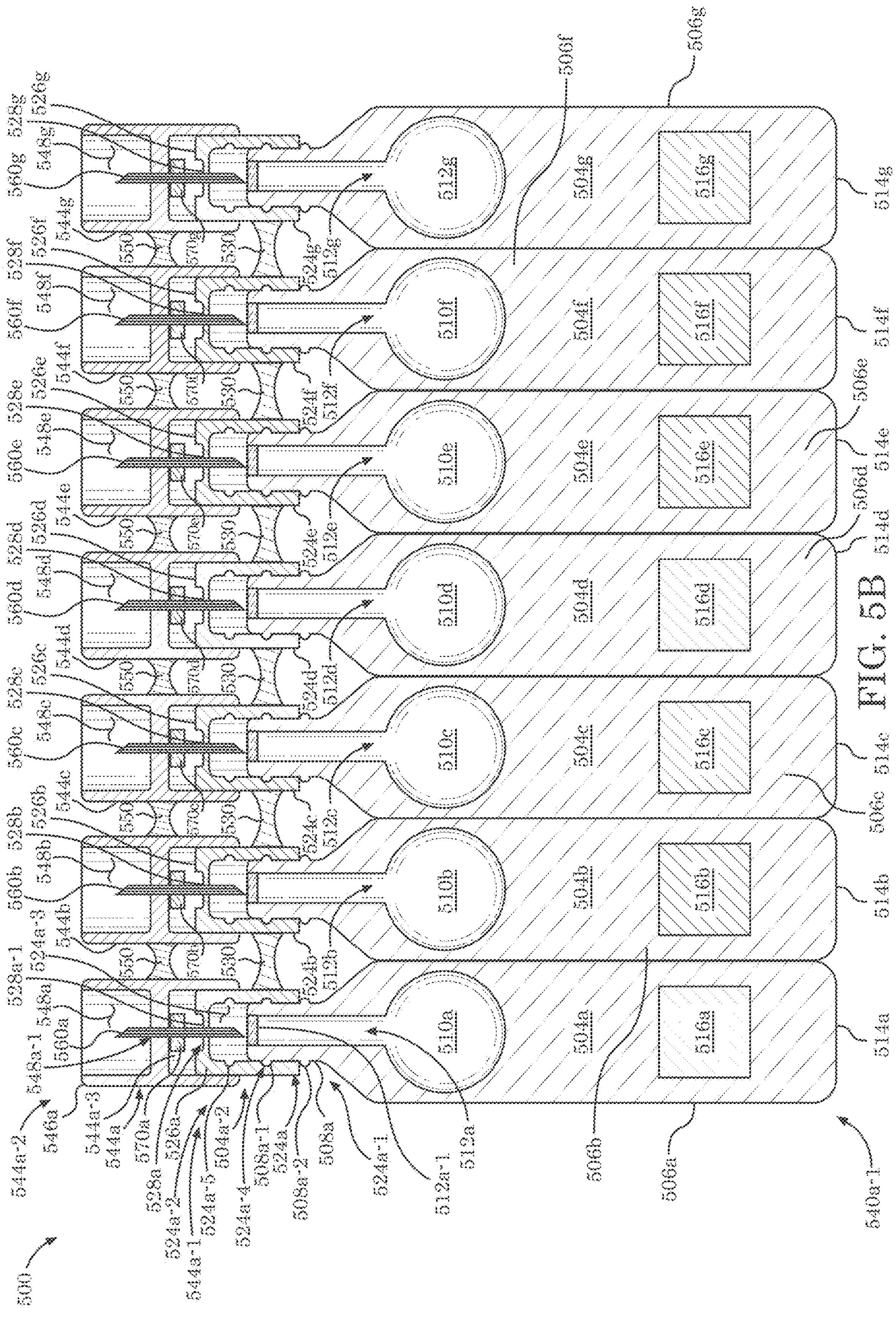
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FIG. 5B

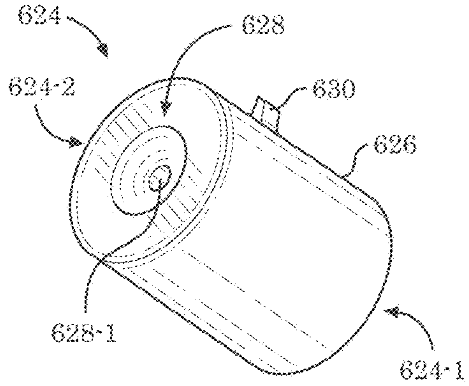

FIG. 6A

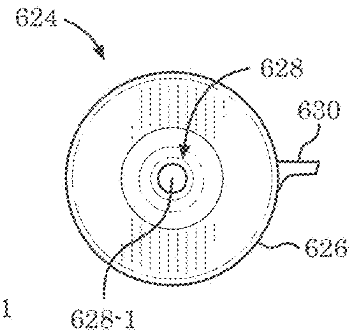

FIG. 6B

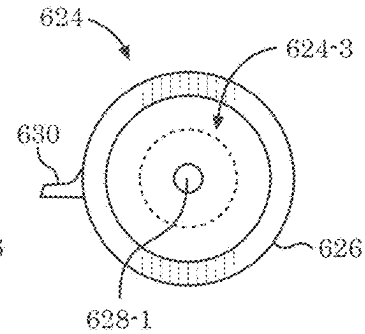

FIG. 6C

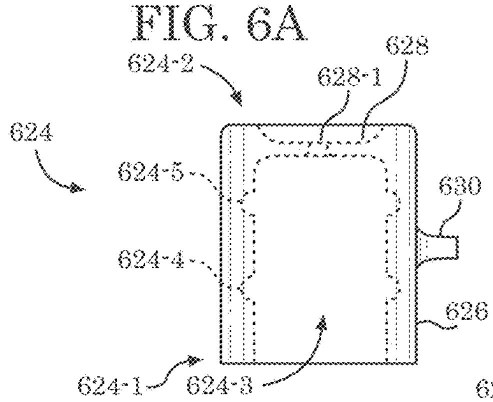

FIG. 6D

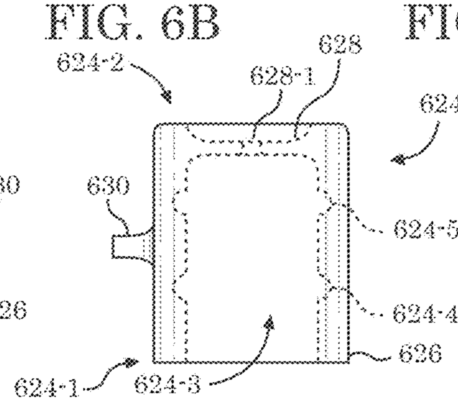

FIG. 6E

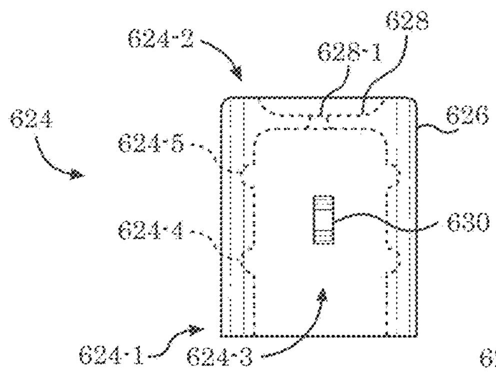

FIG. 6F

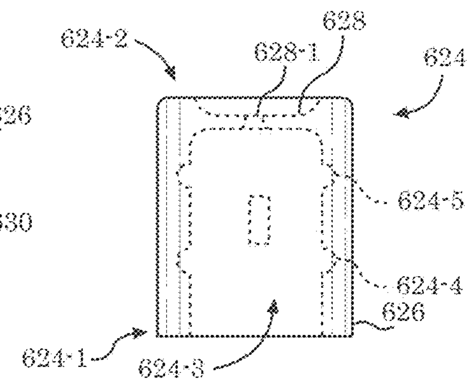

FIG. 6G

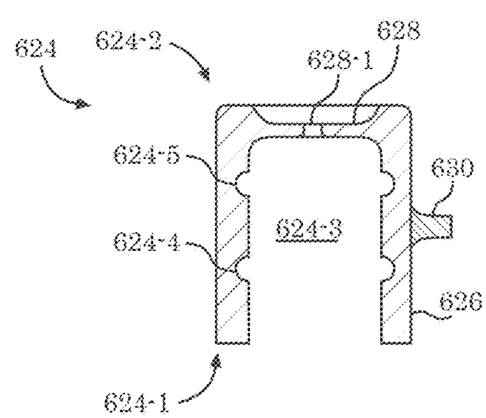

FIG. 6H

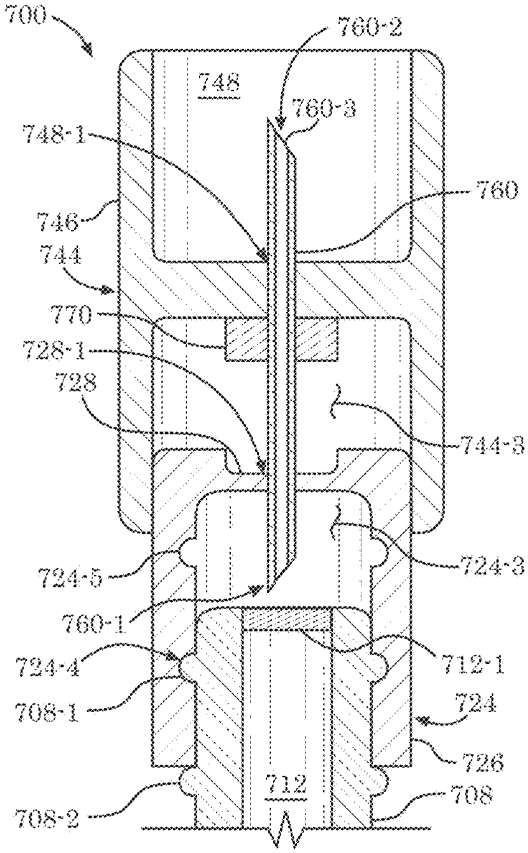

FIG. 7A

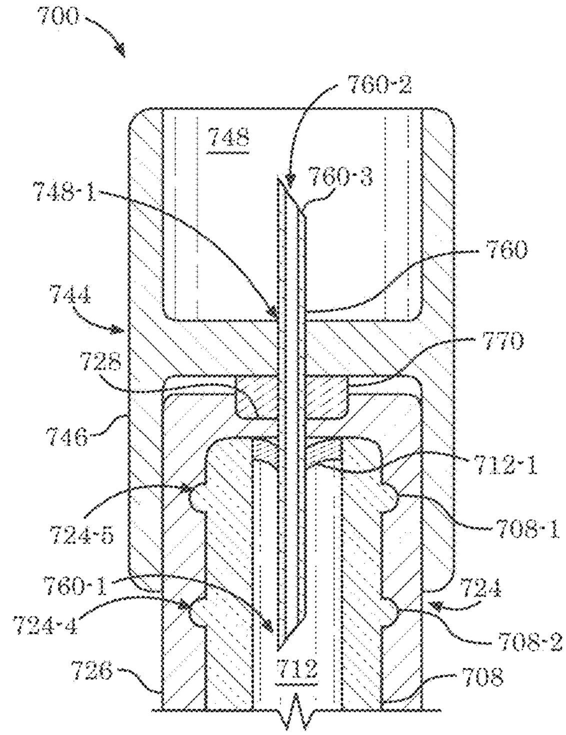

FIG. 7B

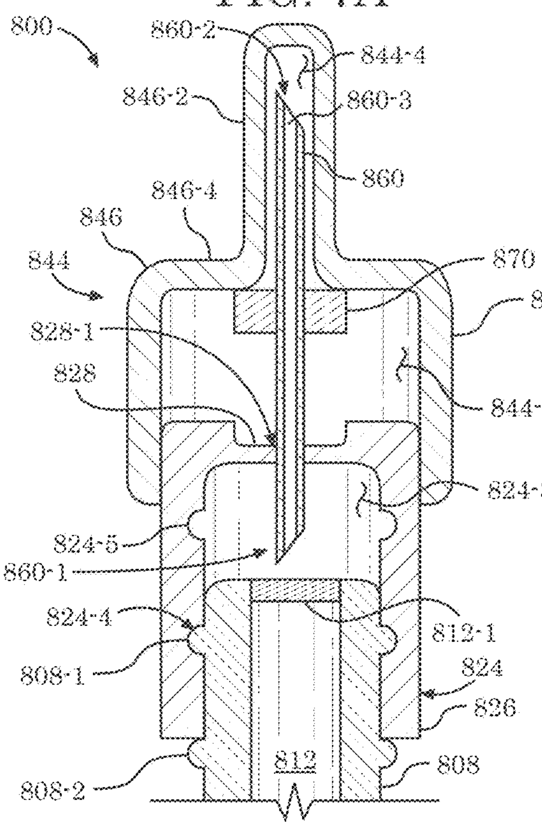

FIG. 8A

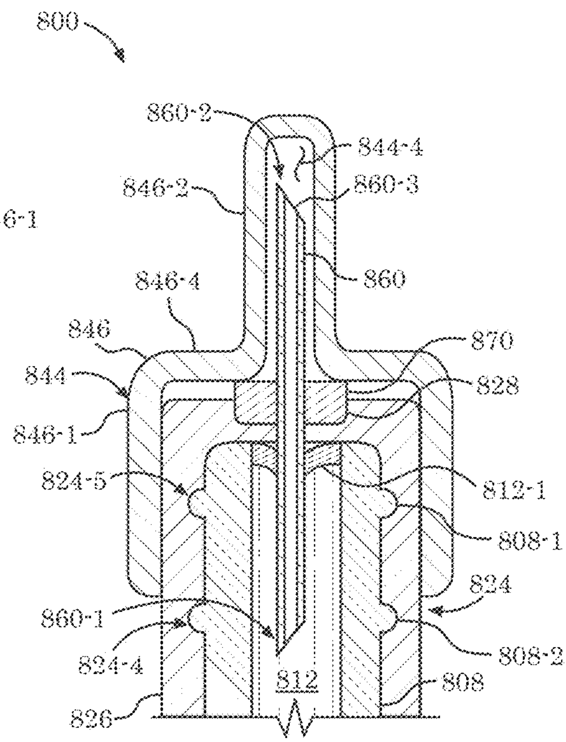

FIG. 8B

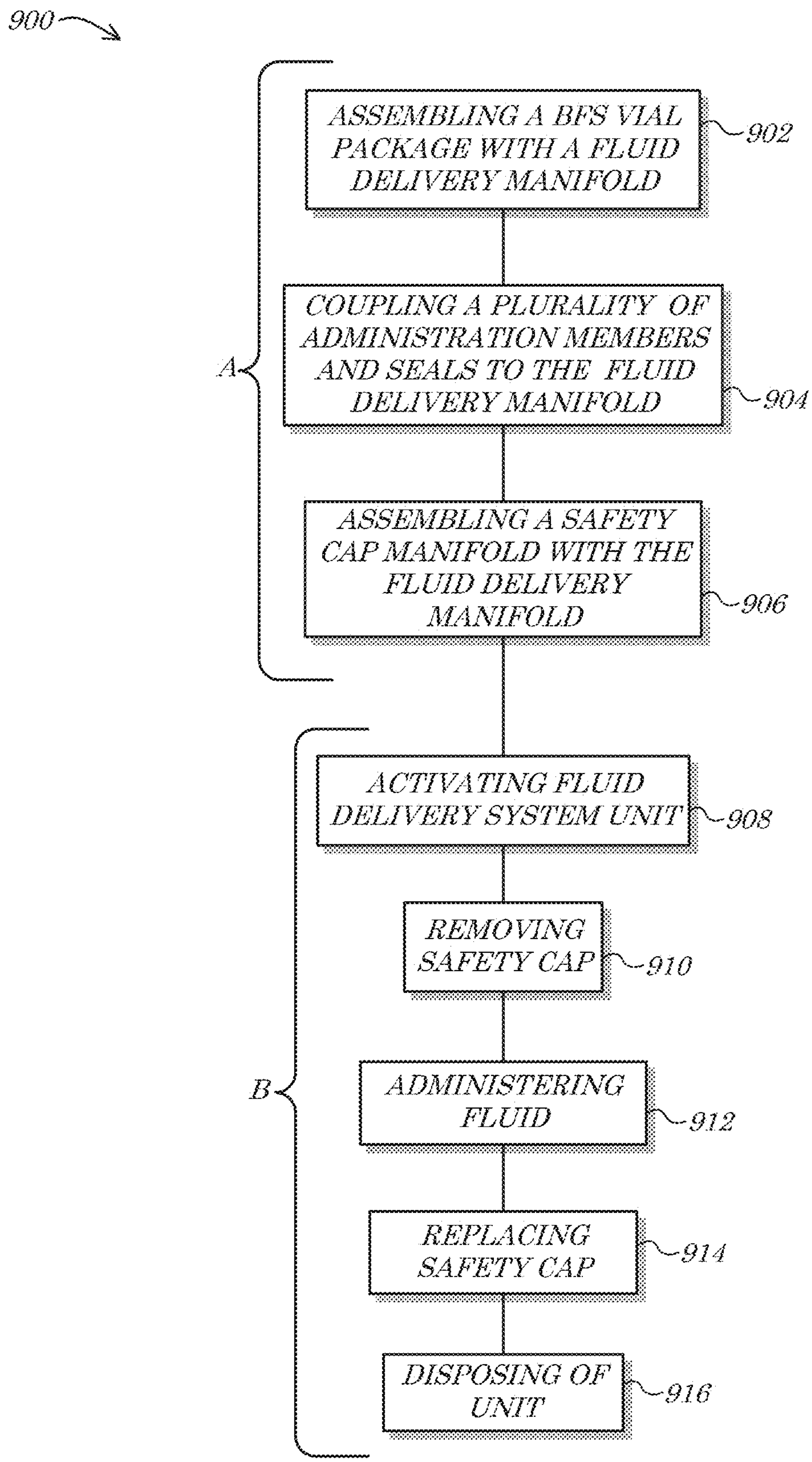
900
ASSEMBLING A BFS VIAL PACKAGE WITH A FLUID DELIVERY MANIFOLD
902
COUPLING A PLURALITY OF ADMINISTRATION MEMBERS AND SEALS TO THE FLUID DELIVERY MANIFOLD
904
ASSEMBLING A SAFETY CAP MANIFOLD WITH THE FLUID DELIVERY MANIFOLD
906
A
ACTIVATING FLUID DELIVERY SYSTEM UNIT
908
REMOVING SAFETY CAP
910
ADMINISTERING FLUID
912
REPLACING SAFETY CAP
914
DISPOSING OF UNIT
916
B

FIG. 9

NFC-ENABLED MEDICAL DEVICE SYSTEM AND PROCESSES FOR GENERATING ASSOCIATED INFORMATION LAYER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims benefit and priority to, and is a Continuation of U.S. patent application Ser. No. 18/181,664 filed on Mar. 10, 2023 and titled "SYSTEMS AND METHODS FOR FLUID DELIVERY MANIFOLDS", which is a Continuation of U.S. patent application Ser. No. 16/876,417 filed on May 18, 2020 and titled "SYSTEMS AND METHODS FOR FLUID DELIVERY MANIFOLDS", which issued as U.S. Pat. No. 11,607,369 on Mar. 21, 2023 and which itself claims benefit and priority to and is a Continuation of International Patent Application No. PCT/US18/61696 titled "SYSTEMS AND METHODS FOR FLUID DELIVERY MANIFOLDS" and filed on Nov. 16, 2018 which itself claims priority to and is a non-provisional of: (i) U.S. Provisional Application Ser. No. 62/587,879 titled "DELIVERY SYSTEM" and filed on Nov. 17, 2017, (ii) U.S. Provisional Application Ser. No. 62/674,565 titled "NFC-Enabled Drug Containing System and Associated Information Layer" and filed on May 21, 2018, and (iii) U.S. Provisional Application Ser. No. 62/680,116 titled "NFC-ENABLED DRUG CONTAINING SYSTEM AND ASSOCIATED INFORMATION LAYER" and filed on Jun. 4, 2018; the contents of each of which are hereby incorporated by reference herein in their entirety.

BACKGROUND

[0002] Every year, millions of people become infected and die from a variety of diseases, some of which are vaccine-preventable. Although vaccination has led to a dramatic decline in the number of cases of several infectious diseases, some of these diseases remain quite common. In many instances, large populations of the world, particularly in developing countries, suffer from the spread of vaccine-preventable diseases due to ineffective immunization programs, either because of poor implementation, lack of affordable vaccines, or inadequate devices for administering vaccines, or combinations thereof.

[0003] Some implementations of immunization programs generally include administration of vaccines via a typical reusable syringe. However, in many situations, particularly in developing countries, the administration of vaccines occur outside of a hospital and may be provided by a non-professional, such that injections are given to patients without carefully controlling access to syringes. The use of reusable syringes under those circumstances increases the risk of infection and spread of blood-borne diseases, particularly when syringes, which have been previously used and are no longer sterile, are used to administer subsequent injections. For example, the World Health Organization (WHO) estimates that blood-borne diseases, such as Hepatitis and human immunodeficiency virus (HIV), are being transmitted due to reuse of such syringes, resulting the death of more than one million people each year.

[0004] Previous attempts at providing single-use or disposable injection devices to remedy such problems in the industry have achieved measurable success but have failed to adequately remedy the existing problems. Pre-filled, single-use injection devices manufactured via injection molding or Form-Fill-Seal (FFS) processes, such as the Uniject™ device available from the Becton, Dickinson and Company of Franklin Lakes, NJ, for example, while offering precise manufacturing tolerances in the range of two thousandths of an inch (0.002-in; 50.8 μm) to four thousandths of an inch (0.004-in; 101.6 μm)-for hole diameters in molded parts, require separate sterilization processes (e.g., gamma radiation) that are not compatible with certain fluids, provide production rates limited to approximately nine thousand (9,000) non-sterile units per hour, and can be provided to an end-user for approximately one dollar and forty cents ($1.40) per dose/unit.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] An understanding of embodiments described herein and many of the attendant advantages thereof may be readily obtained by reference to the following detailed description when considered with the accompanying drawings, wherein:

[0006] FIG. **1** is a front view of BFS vial package according to some embodiments;

[0007] FIG. **2** is a front view of a fluid delivery manifold according to some embodiments;

[0008] FIG. **3** is a front view of a safety cap manifold according to some embodiments;

[0009] FIG. **4** is a front view of a safety cap manifold according to some embodiments;

[0010] FIG. **5**A and FIG. **5**B are front and cross-sectional views of a fluid delivery manifold system according to some embodiments;

[0011] FIG. **6**A, FIG. **6**B, FIG. **6**C, FIG. **6**D, FIG. **6**E, FIG. **6**F, FIG. **6**G, and FIG. **6**H are top-front perspective, top, bottom, left, right, front, back, and cross-sectional views of a fluid delivery hub according to some embodiments;

[0012] FIG. **7**A and FIG. **7**B are cross-sectional views of a fluid delivery manifold system according to some embodiments;

[0013] FIG. **8**A and FIG. **8**B are cross-sectional views of a fluid delivery manifold system according to some embodiments; and

[0014] FIG. **9** is a flow diagram of a method according to some embodiments.

DETAILED DESCRIPTION

I. Introduction

[0015] Embodiments of the present invention provide systems and methods for fluid delivery manifolds that overcome the drawbacks of current delivery devices and methods. For example, the delivery system of some embodiments may include a Blow-Fill-Seal (BFS) package coupled to a fluid delivery manifold and/or a safety cap manifold. In some embodiments, an administration member such as a needle may be selectively actuated by application of force to the fluid delivery manifold and/or a safety cap manifold, causing the administration member to pierce a fluid reservoir of at least one BFS vial of the BFS package. Utilization of such systems that employ BFS vials and/or a fluid delivery manifold may be advantageous and may address various shortcomings of pervious systems.

[0016] BFS vials may, for example, offer a less expensive alternative to vials or devices created via other manufacturing techniques. In some embodiments, BFS vials (e.g., due to the nature of the BFS manufacturing process) may not require separate sterilization (e.g., an may accordingly be compatible with a wider array of fluids), may provide enhanced production rates of approximately thirty thousand (30,000) sterile/aseptic units per hour, and/or may be provided to an end-user for approximately forty-five cents ($0.45) per dose/unit. In some embodiments, these advantages may come with an attendant drawback of reduced manufacturing tolerances. BFS processes may, for example, offer less precise manufacturing tolerances in the range of five hundredths of an inch (0.05-in; 1.27 mm) to fifteen hundredths of an inch (0.15-in; 3.81 mm)-for linear dimensions, e.g., in accordance with the standard ISO 2768-1 "General tolerances for linear and angular dimensions without individual tolerance indications" published by the International Organization for Standardization (ISO) of Geneva, Switzerland (Nov. 15, 1989).

[0017] According to some embodiments, a single-dose, disposable, and/or non-refillable fluid delivery device may be provided despite the presence of the less precise manufacturing tolerances of the BFS vials, by utilization of the systems and methods described herein.

II. Fluid Delivery Manifold Systems

[0018] Referring initially to FIG. **1**, for example, a front view of BFS vial package **102** according to some embodiments is shown. The BFS vial package **102** may, for example, comprise and/or define a plurality of individual-dose BFS vials **104***a-g* joined, coupled, and/or formed together, e.g., as depicted. In some embodiments, and as illustrated with respect to a first one of the plurality of BFS vial **104***a*, the first BFS vial **104***a* may comprise and/or define a first or proximate end **104***a*-**1** and a second or distal end **104***a*-**2**. Disposed therebetween, according to some embodiments, may be a first BFS vial body **106***a*. In some embodiments, each BFS vial **104***a-g* may be coupled (e.g., during and/or as a by-product of a BFS formation process) to any adjacent BFS vial **104***a-g* via the respectively-adjacent BFS bodies **106***a-g*. The connected series of seven (7) BFS vials **104***a-g* depicted in FIG. **1** may, for example, be formed together via a single, seven-compartment BFS mold or die (not shown) during a BFS formation process. In some embodiments, fewer or more BFS vials **104***a-g* may comprise and/or define the BFS vial package **102**, as is or becomes known or practicable.

[0019] According to some embodiments, each BFS vial **104***a-g* may comprise and/or define various features such as features molded, formed, cut, glued, and/or otherwise coupled thereto. As depicted in FIG. **1** for example, the first BFS vial **104***a* may comprise a first vial neck **108***a* (e.g., at or near the distal end **104***a*-**2**) upon which various mating features are formed (or otherwise coupled). In some embodiments, the first vial neck **108***a* may comprise a first or distal exterior radial flange **108***a*-**1** and/or a second or proximate exterior radial flange **108***a*-**2**. According to some embodiments, either of the radial flanges **108***a*-**1**, **108***a*-**2** and/or a separate mating feature (not shown) on or of the first vial neck **108***a* may comprise a tab or other radial protrusion that permits indexed mating of the first vial neck **108***a* with various other components (not shown in FIG. **1**). In some embodiments, each of the BFS vials **104***a-g* may comprise and/or define a fluid reservoir **110***a-g*, e.g., disposed and/or formed on each respective vial body **106***a-g* thereof. According to some embodiments, each fluid reservoir **110***a-g* may store, house, and/or accept a single dose of fluid such as one or more medications, vaccines, solutions, and/or other therapeutic, restorative, preventative and/or curative agents. In some embodiments, a nitrogen bubble (not shown) may be disposed in the fluid reservoirs **110***a-g* to facilitate expelling of all of the fluid in the case that the fluid reservoirs **110***a-g* are squeezed and/or compressed by a user (not shown).

[0020] Radially inward compressive force applied to a first fluid reservoir **110***a* of the first BFS vial **104***a* may, for example, cause the first fluid reservoir **110***a* to collapse radially inward, forcing the fluid stored therein into a fluid channel **112***a* disposed in the first vial neck **108***a*. The first vial neck **108***a* may comprise a tube or passage, for example, that but for a fluid seal **112***a*-**1** disposed at the distal end **104***a*-**2**, would form an opening at the distal end **104***a*-**2** through which the fluid may flow when expelled from the first fluid reservoir **110***a*. In some embodiments, the first vial body **106***a* may comprise and/or define a flat or grip portion **114***a* disposed between the first fluid reservoir **110***a* and the proximate end **104***a*-**1** of the first BFS vial **104***a*. The grip portion **114***a* may, for example, comprise a flat element that permits axial force to be applied to the first BFS vial **104***a* without causing such axial force to be applied to the first fluid reservoir **110***a* (e.g., for engaging an administering element (not shown) as described herein).

[0021] In some embodiments, the BFS vial package **102** may comprise an indicia imprinting thereon (and/or therein) and/or upon each individual BFS vial **104***a-g* (e.g., on the grip portion **114***a* of the first BFS vial **104***a*). Exemplary indicia may include, but is not limited to, lot number, expiration date, medication information (e.g., type, quantity, etc.), a security stamp (color changing temperature sensor to provide indication of whether BFS vials **104***a-g* have or have not been maintained at required temperature), as well as a dosage and/or measurement line provided on each BF vial **104***a-g*. While seven (**7**) BFS vials **104***a-g* are depicted in FIG. **1** as being coupled to and/or comprising the BFS vial package **102**, fewer or more BFS vials **104***a-g* may be coupled to the BFS vial package **102** as is or becomes desirable and/or practicable.

[0022] According to some embodiments, the BFS vial package **102** and/or one or more of the BFS vials **104***a-g* may comprise and/or be coupled to one or more electronic devices **116***a-g*. The elcetronic devices **116***a-g* may comprise, for example, one or more passive inductive, Radio Frequency IDentification (RFID), Near-Field-Communication (NFC), processing, power storage, and/or memory storage devices. In some embodiments, the electronic devices **116***a-g* may store, process, receive, and/or transmit various data elements such as to track geographical movement of the BFS vial package **102** and/or to verify or confirm that a particular BFS vial **104***a-g* should be utilized for administration to a particular recipient.

[0023] In some embodiments for example, in the case that a user (whether a healthcare worker, a patient who is self-injecting or a friend or family member injecting a patient) is about to inject a fluid agent utilizing the first BFS vial **104***a* that comprises a first electronic device **116***a* such as an NFC chip, the user may first open an appropriate app on his/her mobile device (not shown) and tap (or bring into proximity) the first BFS vial **104***a* and/or the first electronic device **116***a* to the mobile device to initiate communications therebetween. This may allow, in some embodiments, the app to verify the injection by verifying and/or authenticating one or more of the following: (i) that the fluid is the correct fluid agent the patient is supposed to be receiving (e.g., based on a comparison of fluid data stored by the first electronic device **116***a* and patient information stored in a separate memory device, e.g., of the user's mobile device); (ii) that the injection is being administered within an appropriate window of time (e.g., by comparing a current time and/or date to a time and/or date stored and/or referenced by data of the first electronic device **116***a*); (iii) that the first BFS vial **104***a* and/or fluid therein has not been compromised and/or is not expired. According to some embodiments, any of the foregoing may be verified based on records of the patient stored in the app on the mobile device and/or accessible to the app via the internet or a cloud-based system and/or data stored in the first electronic device **116***a*. In some embodiments, in the case that the required (or desired) verifications are processed successfully, the user may be authorized to perform the injection (e.g., an approval indicator may be output to the user via a screen of the app). The user may be motivated to only inject upon receiving the approval via the app because the user may only receive rewards via the app if he/she performs approved injections. In some embodiments, the user may also be requested to upload a photo of the first BFS vial **104***a* and/or the injection site (not shown) after the injection (e.g., to finalize qualification for a reward and/or to qualify for an additional reward). According to some embodiments, a reward may be provided to the user via the user's mobile device, e.g., in response to receiving data descriptive of the administration of the fluid and/or the use of the first BFS vial **104***a*.

[0024] In some embodiments, fewer or more components **104***a-g*, **104***a***-1**, **104***a***-2**, **106***a-g*, **108***a*, **108***a***-1**, **108***a***-2**, **110***a-g*, **112***a*, **112***a***-1**, **114***a-g*, **116***a-g* and/or various configurations of the depicted components **104***a-g*, **104***a***-1**, **104***a***-2**, **106***a-g*, **108***a*, **108***a***-1**, **108***a***-2**, **110***a-g*, **112***a*, **112***a***-1**, **114***a-g*, **116***a-g* may be included in the BFS vial package **102** without deviating from the scope of embodiments described herein. In some embodiments, the components **104***a-g*, **104***a***-1**, **104***a***-2**, **106***a-g*, **108***a*, **108***a***-1**, **108***a***-2**, **110***a-g*, **112***a*, **112***a***-1**, **114***a-g*, **116***a-g* may be similar in configuration and/or functionality to similarly named and/or numbered components as described herein. In some embodiments, the BFS vial package **102** (and/or portion and/or component **104***a-g*, **104***a***-1**, **104***a***-2**, **106***a-g*, **108***a*, **108***a***-1**, **108***a***-2**, **110***a-g*, **112***a*, **112***a***-1**, **114***a-g*, **116***a-g* thereof) may be utilized in accordance with the method **900** of FIG. **9** herein, and/or portions thereof.

[0025] Turning now to FIG. **2**, a front view of a fluid delivery manifold **222** according to some embodiments is shown. The fluid delivery manifold **222** may, for example, be configured to mate with a BFS vial package (not shown; such as the BFS vial package **102** of FIG. **1** herein). In some embodiments, the fluid delivery manifold **222** may comprise a plurality of fluid delivery hubs **224***a-g*, each, e.g., configured to mate with a BFS vial (not shown; such as the BFS vials **104***a-g* of FIG. **1** herein). According to some embodiments, one or more of the fluid delivery hubs **224***a-g*, such as a first one of the fluid delivery hubs **224***a*, may comprise and/or define a first or proximate end **224***a***-1** and a second or distal end **224***a***-2**. In some embodiments, the first fluid delivery hub **224***a* may comprise and/or define a first interior volume **224***a***-3** comprising one or more internal features such as a first or proximate radial channel **224***a***-4** and/or a second or distal radial channel **224***a***-5**. According to some embodiments, the fluid delivery hubs **224***a-g* may comprise hub bodies **226***a-g*, e.g., a first hub body **226***a* of the first fluid delivery hub **224***a* being disposed between the first and second ends **224***a***-1**, **224***a***-2** thereof. In some embodiments, the hub bodies **226***a-g* may be substantially cylindrical.

[0026] According to some embodiments, the hub bodies **226***a-g* may comprise and/or define seats **228***a-g*. As depicted in FIG. **2**, for example, the first hub body **226***a* may comprise a first seat **228***a* disposed or formed on an upper surface at the distal end **224***a***-2** of the first hub body **226***a*. In some embodiments, the first seat **228***a* may be in communication with the first interior volume **224***a***-3** via a bore **228***a***-1** disposed therebetween. The bore **228***a***-1** may, for example, allow for passage of an administration member (not shown) coupled to provide fluid (e.g., toward the distal end **224***a***-2**) from a BFS vial (not shown in FIG. **2**) coupled and/or disposed in the first interior volume **224***a***-3**. According to some embodiments, the seats **228***a-g* may be sized and/or configured to accept, couple to, and/or mater with a seal, washer, stopper, and/or other element (not shown) disposed therein.

[0027] In some embodiments, the hub bodies **226***a-g* may be joined together to adjacent hub bodies **226***a-g* via one or more hub connectors **230**. The hub connectors **230** and the hub bodies **226***a-g* may be formed together as a result of a BFS manufacturing process, for example, such as by being extruded from the same plastic and/or polymer material acted upon by a single BFS mold or die. According to some embodiments, the hub connectors **230** may be configured to be easily removed from the hub bodies **226***a-g* such as by incorporating perforations, stress points, and/or break points that are designed to shear, tear, or sever in response to certain applied forces. The first fluid delivery hub **224***a* may be separated from an adjacent second fluid delivery hub **224***b*, for example, in response to a rotational or twisting force being applied to the joint or juncture where a hub connector **230** is coupled to the first hub body **226***a* (and/or a second hub body **226***b*). In some embodiments, a designed stress or break point may be disposed at the juncture with the hub bodies **226***a-g*, in the middle of a length of the hub connectors **230**, or as is or becomes desirable and/or practicable. In such a manner, for example, the fluid delivery manifold **222** may be quickly and easily manufactured as a single part while allowing the individual fluid delivery hubs **224***a-g* to be selectively removed with minimal effort, as is known in the plastic manufacturing arts.

[0028] According to some embodiments, fewer or more components **224***a-g*, **224***a***-1**, **224***a***-2**, **224***a***-3**, **224***a***-4**, **224***a***-5**, **226***a-g*, **228***a-g*, **228***a***-1**, **230** and/or various configurations of the depicted components **224***a-g*, **224***a***-1**, **224***a***-2**, **224***a***-3**, **224***a***-4**, **224***a***-5**, **226***a-g*, **228***a-g*, **228***a***-1**, **230** may be included in the fluid delivery manifold **222** without deviating from the scope of embodiments described herein. In some embodiments, the components **224***a-g*, **224***a***-1**, **224***a***-2**, **224***a***-3**, **224***a***-4**, **224***a***-5**, **226***a-g*, **228***a-g*, **228***a***-1**, **230** may be similar in configuration and/or functionality to similarly named and/or numbered components as described herein. In some embodiments, the fluid delivery manifold **222** (and/or portion and/or component **224***a-g*, **224***a***-1**, **224***a***-2**, **224***a***-3**, **224***a***-4**, **224***a***-5**, **226***a-g*, **228***a-g*, **228***a***-1**, **230** thereof) may be utilized in accordance with the method **900** of FIG. **9** herein, and/or portions thereof.

[0029] Turning now to FIG. **3**, a front view of a safety cap manifold **342** according to some embodiments is shown. In some embodiments, the safety cap manifold **342** may be configured to mate with a fluid delivery manifold (not shown) such as the fluid delivery manifold **222** of FIG. **2** herein. The safety cap manifold **342** may, for example, comprise a plurality of safety caps **344***a-g* each being configured to mate with a respective fluid delivery hub (not shown) such as the fluid delivery hubs **224***a-g* of FIG. **2** herein. In some embodiments, one or more of the safety caps **344***a-g*, such as a first one of the safety caps **344***a*, may comprise and/or define a first or proximate end **344***a***-1** and a second or distal end **344***a***-2**. In some embodiments, the first safety cap **344***a* may comprise and/or define a first lower void **344***a***-3**. According to some embodiments, the safety caps **344***a-g* may comprise safety cap bodies **346***a-g*, e.g., a first safety cap body **346***a* of the first safety cap **344***a* being disposed between the first and second ends **344***a***-1**, **344***a***-2** thereof. In some embodiments, the safety cap bodies **346***a-g* may be substantially cylindrical.

[0030] According to some embodiments, the safety cap bodies **346***a-g* may comprise and/or define upper voids **348***a-g*. As depicted in FIG. **3**, for example, the first safety cap body **346***a* may comprise a first upper void **348***a* disposed, formed, and/or open at (or closed at) the distal end **344***a***-2** of the first safety cap body **346***a*. In some embodiments, the first upper void **348***a* may be in communication with the first lower void **344***a***-3** via a passage **348***a***-1** disposed therebetween. The passage **348***a***-1** may, for example, allow for passage of an administration member (not shown) housed and/or retained by a fluid delivery hub (not shown) disposed and/or coupled within the first lower void **344***a***-3**.

[0031] In some embodiments, the safety cap bodies **346***a-g* may be joined together to adjacent safety cap bodies **346***a-g* via one or more cap connectors **350**. The cap connectors **350** and the safety cap bodies **346***a-g* may be formed together as a result of a BFS (or other) manufacturing process, for example, such as by being extruded and/or injected from the same plastic and/or polymer material acted upon by a single BFS (or other) mold or die. According to some embodiments, the cap connectors **350** may be configured to be easily removed from the safety cap bodies **346***a-g* such as by incorporating perforations, stress points, and/or break points that are designed to shear, tear, or sever in response to certain applied forces. The first safety cap **344***a* may be separated from an adjacent second safety cap **344***b*, for example, in response to a rotational or twisting force being applied to the joint or juncture where a cap connector **350** is coupled to the first safety cap body **346***a* (and/or a second safety cap body **346***b*). In some embodiments, a designed stress or break point may be disposed at the juncture with the safety cap bodies **346***a-g*, in the middle of a length of the cap connectors **350**, or as is or becomes desirable and/or practicable. In such a manner, for example, the safety cap manifold **342** may be quickly and easily manufactured as a single part while allowing the individual safety caps **344***a-g* to be selectively removed with minimal effort, as is known in the plastic manufacturing arts.

[0032] According to some embodiments, fewer or more components **344***a-g*, **344***a***-1**, **344***a***-2**, **344***a***-3**, **346***a-g*, **348***a-g*, **348***a***-1**, **350** and/or various configurations of the depicted components **344***a-g*, **344***a***-1**, **344***a***-2**, **344***a***-3**, **346***a-g*, **348***a-g*, **348***a***-1**, **350** may be included in the safety cap manifold **342** without deviating from the scope of embodiments described herein. In some embodiments, the components **344***a-g*, **344***a***-1**, **344***a***-2**, **344***a***-3**, **346***a-g*, **348***a-g*, **348***a***-1**, **350** may be similar in configuration and/or functionality to similarly named and/or numbered components as described herein. In some embodiments, the safety cap manifold **342** (and/or portion and/or component **344***a-g*, **344***a***-1**, **344***a***-2**, **344***a***-3**, **346***a-g*, **348***a-g*, **348***a***-1**, **350** thereof) may be utilized in accordance with the methods **900** of FIG. **9** herein, and/or portions thereof.

[0033] Referring now to FIG. **4**, a front view of a safety cap manifold **442** according to some embodiments is shown. In some embodiments, the safety cap manifold **442** may be configured to mate with a fluid delivery manifold (not shown) such as the fluid delivery manifold **222** of FIG. **2** herein. The safety cap manifold **442** may, for example, comprise a plurality of safety caps **444***a-g* each being configured to mate with a respective fluid delivery hub (not shown) such as the fluid delivery hubs **224***a-g* of FIG. **2** herein. In some embodiments, one or more of the safety caps **444***a-g*, such as a first one of the safety caps **444***a*, may comprise and/or define a first or proximate end **444***a***-1** and a second or distal end **444***a***-2**. In some embodiments, the first safety cap **444***a* may comprise and/or define a first lower void **444***a***-3** (e.g., having a first diameter) and/or a first upper void **444***a***-4** (e.g., having a second diameter) in communication therewith. According to some embodiments, the safety caps **444***a-g* may comprise safety cap bodies **446***a-g*, e.g., a first safety cap body **446***a* of the first safety cap **444***a* being disposed between the first and second ends **444***a***-1**, **444***a***-2** thereof. In some embodiments, the safety cap bodies **446***a-g* may be conically shaped and/or may comprise a plurality of shaped portions.

[0034] As depicted in FIG. **4** for example, the first safety cap body **446***a* may comprise a lower body portion **446***a***-1** that is cylindrically-shaped with a first outside diameter and an upper body portion **446***a***-2** that is cylindrically-shaped with a second outside diameter that is smaller that the first outside diameter. According to some embodiments, a smooth transition of the outside surface of the first safety cap body **446***a* between the first and second outside diameters may define a tapered body portion **446***a***-3** between the lower body portion **446***a***-1** and the upper body portion **446***a***-1**. In some embodiments, the upper body portion **446***a***-2** may be frustoconically-shaped and/or may define therein the upper void **444***a***-4**. The upper void **444***a***-4** and/or the upper body portion **446***a***-2** may, for example, allow for passage of an administration member (not shown) housed and/or retained by a fluid delivery hub (not shown) disposed and/or coupled within the first lower void **444***a***-3**.

[0035] According to some embodiments, the first safety cap body **446***a* may comprise and/or define an exterior flange **446***a***-4**. The exterior flange **446***a***-4** may, for example, be formed and/or defined by the intersection of the upper body portion **446***a***-2** and the lower body portion **446***a***-1** (and/or the tapered body portion **446***a***-3**), e.g., having a radial depth based on the difference between the first and second outside diameters thereof. In some embodiments, the exterior flange **446***a***-4** and/or the tapered body portion **446***a***-3** may be utilized to apply axial forces on the first safety cap **444***a*, e.g., to effectuate activation of a fluid delivery system (not shown) as described herein.

[0036] In some embodiments, the safety cap bodies **446***a-g* may be joined together to adjacent safety cap bodies **446***a-g* via one or more cap connectors **450**. The cap connectors **450** and the safety cap bodies **446***a-g* may be formed together as a result of a BFS (or other) manufacturing process, for example, such as by being extruded and/or injected from the same plastic and/or polymer material acted upon by a single BFS (or other) mold or die. According to some embodiments, the cap connectors **450** may be configured to be easily removed from the safety cap bodies **446***a-g* such as by incorporating perforations, stress points, and/or break points that are designed to shear, tear, or sever in response to certain applied forces. The first safety cap **444***a* may be separated from an adjacent second safety cap **444***b*, for example, in response to a rotational or twisting force being applied to the joint or juncture where a cap connector **450** is coupled to the first safety cap body **446***a* (and/or a second safety cap body **446***b*). In some embodiments, a designed stress or break point may be disposed at the juncture with the safety cap bodies **446***a-g*, in the middle of a length of the cap connectors **450**, or as is or becomes desirable and/or practicable. In such a manner, for example, the safety cap manifold **442** may be quickly and easily manufactured as a single part while allowing the individual safety caps **444***a-g* to be selectively removed with minimal effort, as is known in the plastic manufacturing arts.

[0037] According to some embodiments, fewer or more components **444***a-g*, **444***a*-**1**, **444***a*-**2**, **444***a*-**3**, **444***a*-**4**, **446***a-g*, **446***a*-**1**, **446***a*-**2**, **446***a*-**3**, **446***a*-**4**, **450** and/or various configurations of the depicted components **444***a-g*, **444***a*-**1**, **444***a*-**2**, **444***a*-**3**, **444***a*-**4**, **446***a-g*, **446***a*-**1**, **446***a*-**2**, **446***a*-**3**, **446***a*-**4**, **450** may be included in the safety cap manifold **442** without deviating from the scope of embodiments described herein. In some embodiments, the components **444***a-g*, **444***a*-**1**, **444***a*-**2**, **444***a*-**3**, **444***a*-**4**, **446***a-g*, **446***a*-**1**, **446***a*-**2**, **446***a*-**3**, **446***a*-**4**, **450** may be similar in configuration and/or functionality to similarly named and/or numbered components as described herein. In some embodiments, the safety cap manifold **442** (and/or portion and/or component **444***a-g*, **444***a*-**1**, **444***a*-**2**, **444***a*-**3**, **444***a*-**4**, **446***a-g*, **446***a*-**1**, **446***a*-**2**, **446***a*-**3**, **446***a*-**4**, **450** thereof) may be utilized in accordance with the methods **900** of FIG. **9** herein, and/or portions thereof.

[0038] Turning now to FIG. **5**A and FIG. **5**B, front and cross-sectional views of a fluid delivery manifold system **500** according to some embodiments are shown. The fluid delivery manifold system **500** may comprise, for example, a BFS vial package **502** comprising a plurality if individual BFS vials **504***a-g*. Features of the BFS vials **504***a-g* are described in representative detail by referring to a first one of the BFS vials **504***a* (e.g., the left-most

[0039] BFS vial **504***a-g* of the BFS vial package **502**). According to some embodiments, the first BFS vial **504***a* may comprise a first or proximate end **504***a*-**1** and a second or distal end **504***a*-**2** between which is disposed a first vial body **506***a*. In some embodiments, each BFS vials **504***a-g* may comprise a vial body **506***a-g* that may be coupled to any adjacent vial body **506***a-g* in the BFS vial package **502** and/or may be coupled to and/or define various features. With reference to the first BFS vial **504***a*, for example, the first vial body **506***a* may comprise, define, and/or be coupled to a first vial neck **508***a* having a first or distal radial flange **508***a*-**1** and/or a second or proximate radial flange **508***a*-**2** formed and/or coupled thereto.

[0040] According to some embodiments, each BFS vial **504***a-g* may comprise a fluid reservoir **510***a-g* in communication with a respective fluid passage **512***a-g* (e.g., a first fluid passage **512***a* of the first BFS vial **504***a* being disposed in (or formed by) the first vial neck **508***a*). With reference to the first BFS vial **504***a*, for example, fluid may be stored in (e.g., injected into during an aseptic BFS manufacturing process) a first fluid reservoir **510***a* and/or the first fluid passage **512***a* an may be prevented from existing the first BFS vial **504***a* by a first fluid seal **512***a*-**1** disposed at a distal end of the fluid passage **512***a*. In some embodiments, the BFS vials **504***a-g* may comprise grip portions **514***a-g* and/or electronic devices **516***a-g* (e.g., embedded and/or formed within the BFS vials **504***a-g*).

[0041] In some embodiments, the fluid delivery manifold system **500** may comprise a fluid delivery manifold **522** coupled to the BFS vial package **502**. The fluid delivery manifold **522** may comprise, for example, a plurality of fluid delivery hubs **524***a-g*, each fluid delivery hub **524***a-g* being coupled to a respective one of the BFS vials **504***a-g*. Features of the fluid delivery hubs **524***a-g* are described in representative detail by referring to a first one of the fluid delivery hubs **524***a* (e.g., the left-most fluid delivery hub **524***a-g* of the fluid delivery manifold **522**). The first fluid delivery hub **524***a* may comprise, according to some embodiments, a first or proximate end **524***a*-**1** and a second or distal end **524***a*-**2** and/or may comprise and/or define a first interior volume **524***a*-**3**. In some embodiments, the first interior volume **524***a*-**3** may comprise one or more internal mating features such as a first or proximate radial channel **524***a*-**4** and/or a second or distal radial channel **524***a*-**5**. According to some embodiments, the fluid delivery hubs **524***a-g* may comprise hub bodies **526***a-g*, e.g., a first hub body **526***a* of the first fluid delivery hub **524***a* being disposed between the first and second ends **524***a*-**1**, **524***a*-**2** thereof. In some embodiments, the hub bodies **526***a-g* may be substantially cylindrical.

[0042] As shown in the cross-section of FIG. **5**B, the first vial neck **508***a* may be inserted into the first interior volume **524***a*-**3** by an amount that aligns first radial flange **508***a*-**1** with the proximate radial channel **524***a*-**4**. In some embodiments, engagement and/or insertion of the first vial neck **508***a* into the first interior volume **524***a*-**3** may cause the first vial neck **508***a* of the first BFS vial **504***a* (and/or the first radial flange **508***a*-**1** thereof) to compress radially inward and/or may cause the first hub body **526***a* to expand radially outward (e.g., elastically), to allow continued advancement of the first BFS vial **504***a* into the first interior volume **524***a*-**3**, e.g., in accordance with a tight fit and/or interference fit engagement. According to some embodiments, the radial pressure exerted by forcing the first vial neck **508***a* of the first BFS vial **504***a* into the first interior volume **524***a*-**3** may impart a radial spring effect to the first radial flange **508***a*-**1** (or a portion thereof) such that when axial advancement aligns the first radial flange **508***a*-**1** with the corresponding proximate radial channel **524***a*-**4**, the first radial flange **508***a*-**1** (or a portion thereof) may spring radially outward and into the corresponding proximate radial channel **524***a*-**4**, thereby reducing and/or removing the radial pressure exerted thereon by the difference between an diameter of the first interior volume **524***a*-**3** and an outside diameter and/or radial extents of the first radial flange **508***a*-**1**. In such a manner, for example, each BFS vial **504***a-g* may be snapped into each respective fluid delivery hub **524***a-g*.

[0043] According to some embodiments, the hub bodies **526***a-g* may comprise and/or define seats **528***a-g*. As depicted in FIG. **5**B, for example, the first hub body **526***a* may comprise a first seat **528***a* disposed or formed on an upper surface at the distal end **524***a***-2** of the first hub body **526***a*. In some embodiments, the first seat **528***a* may be in communication with the first interior volume **524***a***-3** via a bore **528***a***-1** disposed therebetween. In some embodiments, the hub bodies **526***a-g* may be joined together to adjacent hub bodies **526***a-g* via one or more hub connectors **530**. In such a manner, for example, the fluid delivery manifold **522** may be mated with or coupled to the BFS vial package **502** (e.g., as part of a manufacturing assembly process).

[0044] In some embodiments, the fluid delivery manifold system **500** may comprise a safety cap manifold **542** coupled to the fluid delivery manifold **522**. The safety cap manifold **542** may comprise, for example, a plurality of safety caps **544***a-g*, each safety cap **544***a-g* being coupled to a respective one of the fluid delivery hubs **524***a-g*. Features of the safety caps **544***a-g* are described in representative detail by referring to a first one of the safety caps **544***a* (e.g., the left-most safety cap **544***a-g* of the safety cap manifold **542**). The first safety cap **544***a* may comprise, according to some embodiments, a first or proximate end **544***a***-1** and a second or distal end **544***a***-2** and/or may comprise and/or define a lower void **544***a***-3**, e.g., open at the proximate end **544***a***-1**. In some embodiments, the lower void **544***a***-3** may comprise one or more internal mating features (not shown) such as to mate to corresponding features (also not shown) on an outside surface of the first fluid delivery hub **524***a* to which the first safety cap **544***a* is coupled.

[0045] According to some embodiments, the first safety cap **544***a* may comprise a first cap body **546***a* disposed between the first and second ends **544***a***-1**, **544***a***-2** thereof. In some embodiments, the first cap body **546***a* may be substantially cylindrical. In some embodiments, the safety caps **544***a-g* may comprise and/or define upper voids **548***a-g*. The first safety cap **544***a* may comprise a first upper void **548***a* open at the distal end **544***a***-2**, for example, and/or in communication with the first lower void **544***a***-3**, e.g., via a first passage **548***a***-1**. In some embodiments, the cap bodies **546***a-g* may be joined together to adjacent cap bodies **546***a-g* via one or more cap connectors **550**. In such a manner, for example, the safety cap manifold **542** may be mated with or coupled to the fluid delivery manifold **522** (e.g., as part of a manufacturing assembly process).

[0046] In some embodiments, the mated safety caps **544***a-g* and fluid delivery hubs **524***a-g* may house and/or retain needles or administration members **560***a-g* (e.g., for at least one of subcutaneous, intramuscular, intradermal, and intravenous injection) and/or seals **570***a-g*. As shown with respect to the first BFS vial **504***a*, the first fluid delivery hub **524***a*, and the first safety cap **544***a* coupling, for example, a first administration member **560***a* may extend from within the interior volume **524***a***-3** (e.g., proximate to the first fluid seal **512***a***-1**) of the first fluid delivery hub **524***a*, through the bore **528***a***-1** and the first seat **528***a* and into the lower void **544***a***-3** of the first safety cap **544***a*. According to some embodiments, within the lower void **544***a***-3** of the first safety cap **544***a* the first administration member **560***a* may pass through and/or be coupled to a first seal **570** (e.g., an annular rubber seal) and pass through the passage **548***a***-1** of the first safety cap **544***a* and into the first upper void **548***a* thereof. In some embodiments, the upper void **548***a* may be closed at the distal end **544***a***-2**, e.g., to prevent accidental access to the first administration member **560***a*. According to some embodiments, the axial length of the first safety cap **544***a* and/or the axial depth or extent of the first upper void **548***a* may be substantially longer than the length of the first administration member **560***a* exposed in the first upper void **548***a* and/or the upper void **548***a* may otherwise not need to be closed at the distal end **544***a***-2**.

[0047] According to some embodiments, any coupled set of BFS vials **504***a-g*, fluid delivery hubs **524***a-g*, and safety caps **544***a-g* may be configured to be easily removed from the fluid delivery manifold system **500** such as by incorporating perforations, stress points, and/or break points that are designed to shear, tear, or sever in response to certain applied forces. Rotational or twisting force applied to the coupled first BFS vial **504***a*, first fluid delivery hub **524***a*, and first safety cap **544***a* may, for example, generate stresses (i) along a coupling between the first vial body **506***a* and an adjacent second vial body **506***b*, (ii) within the hub connector **530** connecting the first fluid delivery hub **524***a* and an adjacent second delivery hub **524***b*, and/or (iii) within the cap connector **550** connecting the first safety cap **544***a* and an adjacent second safety cap **544***b*. Reduced cross-sectional area at one or more of these various connections may be designed to fail in response to experiencing such stresses, thereby permitting a single-dose BFS vial **504***a-g*, fluid delivery hub **542***a-g*, and safety cap **544***a-g* unit to be easily and selectively removed from the fluid delivery manifold system **500**.

[0048] While a certain style of safety cap **544***a-g* is depicted in FIG. **5**A and FIG. **5**B, it should be understood that different styles and/or configurations of safety caps **544***a-g* (e.g., as described herein) may be utilized without deviating from the scope of some embodiments. According to some embodiments, the fluid delivery manifold system **500** may be depicted in FIG. **5**A and FIG. **5**B in a first, assembled, and/or non-activated state. As depicted, for example, the administration members **560***a-g* have not yet been engaged to puncture the BFS vials **504***a-g* to permit release and/or administration of the fluid stored in the fluid reservoirs **510***a-g*. In some embodiments, an axial compressive force may be applied to the safety caps **544***a-g* and/or the fluid delivery hubs **524***a-g* in one direction and the grip portions **514***a-g* in an opposite direction, to urge the BFS vials **504***a-g* further into the fluid delivery hubs **524***a-g* and cause the administration members **560***a-g* to pierce the respective BFS vials **504***a-g*. In some embodiments, such an activation may be accomplished for single BFS vial **504***a-g* at a time or activation may be effectuated for multiple BFS vials **504***a-g* at one time, e.g., as permitted by the structural configurations, alignment, and engagement of the BFS vial package **502**, the fluid delivery manifold **522**, and the safety cap manifold **542**.

[0049] In some embodiments, fewer or more components **502**, **504***a-g*, **504***a***-1**, **504***a***-2**, **506***a-g*, **508***a*, **508***a***-1**, **508***a***-2**, **510***a-g*, **512***a-g*, **514***a-g*, **516***a-g*, **522**, **524***a-g*, **524***a***-1**, **524***a***-2**, **524***a***-3**, **524***a***-4**, **524***a***-5**, **526***a-g*, **528***a-g*, **528***a***-1**, **530**, **542**, **544***a-g*, **544***a***-1**, **544***a***-2**, **544***a***-3**, **546***a*, **548***a-g*, **548***a***-1**, **550**, **560***a-g*, **570***a-g* and/or various configurations of the depicted components **502**, **504***a-g*, **504***a***-1**, **504***a***-2**, **506***a-g*, **508***a*, **508***a***-1**, **508***a***-2**, **510***a-g*, **512***a-g*, **514***a-g*, **516***a-g*, **522**, **524***a-g*, **524***a***-1**, **524***a***-2**, **524***a***-3**, **524***a***-4**, **524***a***-5**, **526***a-g*, **528***a-g*, **528***a***-1**, **530**, **542**, **544***a-g*, **544***a***-1**, **544***a***-2**, **544***a***-3**, **546***a*, **548***a-g*, **548***a***-1**, **550**, **560***a-g*, **570***a-g* may be included in the fluid delivery manifold system **500** without deviating from the scope of embodiments described herein. In some embodiments, the components **502**, **504***a-g*, **504***a***-1**, **504***a***-2**,

506_a-g_, **508**_a_, **508**_a_-**1**, **508**_a_-**2**, **510**_a-g_, **512**_a-g_, **514**_a-g_, **516**_a-g_, **522**, **524**_a-g_, **524**_a_-**1**, **524**_a_-**2**, **524**_a_-**3**, **524**_a_-**4**, **524**_a_-**5**, **526**_a-g_, **528**_a-g_, **528**_a_-**1**, **530**, **542**, **544**_a-g_, **544**_a_-**1**, **544**_a_-**2**, **544**_a_-**3**, **546**_a_, **548**_a-g_, **548**_a_-**1**, **550**, **560**_a-g_, **570**_a-g_ may be similar in configuration and/or functionality to similarly named and/or numbered components as described herein. In some embodiments, the fluid delivery manifold system **500** (and/or portion and/or component **502**, **504**_a-g_, **504**_a_-**1**, **504**_a_-**2**, **506**_a-g_, **508**_a_, **508**_a_-**1**, **508**_a_-**2**, **510**_a-g_, **512**_a-g_, **514**_a-g_, **516**_a-g_, **522**, **524**_a-g_, **524**_a_-**1**, **524**_a_-**2**, **524**_a_-**3**, **524**_a_-**4**, **524**_a_-**5**, **526**_a-g_, **528**_a-g_, **528**_a_-**1**, **530**, **542**, **544**_a-g_, **544**_a_-**1**, **544**_a_-**2**, **544**_a_-**3**, **546**_a_, **548**_a-g_, **548**_a_-**1**, **550**, **560**_a-g_, **570**_a-g_ thereof) may be utilized in accordance with the method **900** of FIG. **9** herein, and/or portions thereof.

[0050] Turning now to FIG. **6**A, FIG. **6**B, FIG. **6**C, FIG. **6**D, FIG. **6**E, FIG. **6**F, FIG. **6**G, and FIG. **6**H, top-front perspective, top, bottom, left, right, front, back, and cross-sectional views of a fluid delivery hub **624** according to some embodiments are shown. The fluid delivery hub **624** may comprise, for example, a first or proximate end **624-1** and a second or distal end **624-2** and/or may define an interior volume **624-3** (e.g., open at the proximate end **624-1**). In some embodiments, the interior volume **624-3** may comprise and/or define one or more mating features (e.g., on an inside surface of the fluid delivery hub **624**) such as a first or proximate radial channel **624-4** and/or a second or distal radial channel **624-5**. According to some embodiments, the channels **624-4**, **624-5** may be beveled and/or rounded, as shown, to facilitate entry and exit of corresponding mating flanges (not shown) thereof. In some embodiments, a lower edge of one or more of the channels **624-4**, **624-5** may comprise a right-angle or other angular profile (not shown) such that a corresponding mating feature may readily enter the channel **624-4**, **624-5** from the proximate end **624-1** but may be impeded from exiting the channel **624-4**, **624-5** (e.g., from backing out) toward the proximate end **624-1**. In such a manner, for example, a BFS vial (not shown) with a corresponding mating feature(s) may be easily inserted into the interior volume **624-3** but may be locked therein once engaged, e.g., to prevent attempted reuse of the fluid delivery hub **624**.

[0051] According to some embodiments, the interior volume **624-3** may be defined by a hub body **626** such as a cylindrical shape as-depicted. In some embodiments, the fluid delivery hub **624** (and/or the hub body **626** thereof) may comprise and/or define a seat **628**. The seat **628** may comprise, for example, a depression, indent, and/or countersink feature disposed on a top surface of the hub body **626** at the distal end **624-2**. In some embodiments, the seat **628** may comprise a void or volume that is in communication with the interior volume **624-3** via a bore **628-1**. According to some embodiments, the fluid delivery hub **624** (and/or the hub body **626** thereof) may comprise a hub connector **630** that may, for example, comprise an element molded and/or coupled to the hub body **626** and operable to be coupled to an adjacent fluid delivery hub (not shown). In some embodiments, the hub connector **630** may comprise a portion of a connective element that connected the hub body **626** to and adjacent hub body and then was selectively severed to remove the fluid delivery hub **624**, e.g., from a fluid delivery manifold (not shown) such as the fluid delivery manifolds **222**, **522**, of FIG. **2**, FIG. **5**A, and/or FIG. **5**B herein.

[0052] In some embodiments, fewer or more components **624-1**, **624-2**, **624-3**, **624-4**, **624-5**, **626**, **628**, **628-1**, **630** and/or various configurations of the depicted components **624-1**, **624-2**, **624-3**, **624-4**, **624-5**, **626**, **628**, **628-1**, **630** may be included in the fluid delivery hub **624** without deviating from the scope of embodiments described herein. In some embodiments, the components **624-1**, **624-2**, **624-3**, **624-4**, **624-5**, **626**, **628**, **628-1**, **630** may be similar in configuration and/or functionality to similarly named and/or numbered components as described herein. In some embodiments, the fluid delivery hub **624** (and/or portion and/or component **624-1**, **624-2**, **624-3**, **624-4**, **624-5**, **626**, **628**, **628-1**, **630** thereof) may be utilized in accordance with the method **900** of FIG. **9** herein, and/or portions thereof.

[0053] Turning now to FIG. **7**A and FIG. **7**B, cross-sectional views of a fluid delivery manifold system **700** according to some embodiments are shown. In some embodiments, the fluid delivery manifold system **700** may be depicted in various states of activation in FIG. **7**A and FIG. **7**B. In FIG. **7**A, for example, the fluid delivery manifold system **700** may be depicted in a first or unengaged state while in FIG. **7**B the fluid delivery manifold system **700** may be depicted in a second or engaged state. According to some embodiments, a user (not shown) may progress the fluid delivery manifold system **700** from the first state to the second state by selective application of axial force that ultimately causes a fluid to be delivered to a recipient (not shown; e.g., a therapeutic and/or medical fluid to be delivered to a patient).

[0054] According to some embodiments, the fluid delivery manifold system **700** may comprise a BFS vial neck **708** having a first or distal exterior radial flange **708-1** and a second or proximate exterior radial flange **708-2**. In some embodiments, a fluid (not explicitly shown) may reside in and/or travel through a fluid passage **712** within the BFS vial neck **708** and/or may be contained by a fluid seal **712-1**. The fluid seal **712-1** may be integral to the BFS vial neck **708** such as by being formed during a BFS manufacturing process together with the BFS vial neck **708** or may comprise a separate element coupled to seal the BFS vial neck **708** and attendant fluid passage **712** thereof.

[0055] In some embodiments, the fluid delivery manifold system **700** may comprise a fluid delivery hub **724** coupled to the BFS vial neck **708**. The BFS vial neck **708** may, for example, be at least partially inserted into an interior volume **724-3** defined by the fluid delivery hub **724**, e.g., as shown in FIG. **7**A. According to some embodiments, the BFS vial neck **708** may have been inserted to a first position within the interior volume **724-3** as shown in FIG. **7**A, such that the first radial flange **708-1** engages with and/or becomes seated in a first or proximate radial channel **724-4** disposed on an inside surface of the interior volume **724-3**. As depicted in FIG. **7**B, continued advancement of the BFS vial neck **708** into the interior volume **724-3** may cause the first radial flange **708-1** to disengage with and/or move out of the first radial channel **724-4** and travel along the inside surface of the interior volume **724-3** until it becomes engaged with and/or seated in a second or distal radial channel **724-5** (as shown in FIG. **7**B). Upon achieving such a second and/or additional level of insertion into the interior volume **724-3**, in some embodiments the second radial flange **708-2** may become engaged with and/or seated in the first radial channel **724-4** (also as shown in FIG. **7**B). In such a manner, for example, at the first state shown in FIG. **7**A the BFS vial neck **708** may be secured to the fluid delivery hub **724** at a first position (e.g., via engagement of the first radial flange

708-1 with the first radial channel **724-4**; e.g., a transport, assembly, and/or pre-engagement position) while at the second state shown in FIG. **7**B the BFS vial neck **708** may be secured to the fluid delivery hub **724** at a second position (e.g., via engagement of the first radial flange **708-1** with the second radial channel **724-5** and engagement of the second radial flange **708-2** with the first radial channel **724-4**; e.g., an engagement position). In some embodiments, the interior volume **724-3** (and/or the channels **724-4**, **724-5** thereof) may be formed and/or defined by a hub body **726**. According to some embodiments, the hub body **726** may comprise and/or define a seat **728** disposed opposite to the interior volume **724-3** and/or a bore **728-1** extending between the seat **728** and the interior volume **724-3**.

[0056] In some embodiments, the fluid delivery manifold system **700** may comprise a safety cap **744** defining a lower void **744-3** into which the hub body **726** is at least partially inserted and/or disposed (e.g., as shown in FIG. **7**A). The lower void **744-3** may comprise an inside diameter equivalent to (or slightly smaller than) an outside diameter of the hub body **726**, for example, such as permitting for a desired fit therebetween. According to some embodiments, the lower void **744-3** may be defined by a cap body **746** that may, for example, be cylindrically-shaped. In some embodiments, the safety cap **744** (and/or cap body **746** thereof) may comprise and/or define an upper void **748** in communication with the lower void **744-3** via a passage **748-1**.

[0057] According to some embodiments, the fluid delivery manifold system **700** may comprise an administration member **760** (such as a needle or tube; e.g., a needle having a length in the ranges of (i) 0.5 mm to 4 mm, (ii) 4mm to 15 mm, or (iii) 15 mm to 30 mm) having a first or proximate end **760-1** and a second or distal end **760-2**. In some embodiments, the administration member **760** may be hollow and/or may otherwise define a fluid channel **760-3** extending from the first end **760-1** to the second end **760-3**. In some embodiments, the administration member **760** may pass through and/or be coupled to a seal **770** disposed in the lower void **744-3**. The seal **770** may comprise, for example, an annular rubber or thermoplastic element that may be deformable or pliable. According to some embodiments, either or both of the first end **760-1** and the second end **760-2** of the administration member **760** may comprise a tip, point, prong, blade, and/or other feature and/or configuration, such as to pierce the fluid seal **712-1** (e.g., in the case of the fires end **760-1**) or pierce an administration surface (such as skin; not shown; e.g., in the case of the second end **760-2**).

[0058] In some embodiments, and as depicted in FIG. **7**A, at the first or disengaged state of the fluid delivery manifold system **700** the first end **760-1** of the administration member **760** may be disposed in the interior volume **724-3** of the fluid delivery hub **724** and the second end **760-2** of the administration member **760** may be disposed in the upper void **748** of the safety cap **744**. The administration member **760** may pass through, for example, each of the bore **728-1** (and the seat **728**), the seal **770**, and the passage **748-1**. According to some embodiments, axial force may be applied to transition the fluid delivery manifold system **700** to the second or engaged state of FIG. **7**B where the first end **760-1** of the administration member **760** has pierced the fluid seal **712-1** and is disposed within the fluid passage **712** of the BFS vial neck **708** and where the seal **770** is seated in, engaged with, and/or coupled to the seat **728**. In such a manner, for example, the fluid delivery manifold system **700** may be engaged to provide the fluid from the fluid passage **712**, through the fluid channel **760-3**, and into a recipient (not shown). In some embodiments, the safety cap **744** may be removed to expose the administration member **760** and/or the second end **760-2** thereof for administration of the fluid. According to some embodiments, axial force (downward, as-oriented in FIG. **7**A and FIG. **7**B) may cause the seal **770** to deform and/or compress into the seat **728**. Axial force applied to the safety cap **744** may cause the cap body **746** (and/or the ceiling of the interior volume **724-3**) to act upon the seal **770** and urge the seal **770** axially downward (e.g., as depicted transitioning from FIG. **7**A to FIG. **7**B). In some embodiments, once the seal **770** contacts the seat **728** it may be forced therein and/or may be deformed or compressed into the seat **728**.

[0059] According to some embodiments, upon the seal **770** achieving a degree of compression and/or deformation, force may be transferred to the hub body **726**, causing the first radial flange **708-1** of the BFS vial neck **708** to unseat and/or disengage from the first radial channel **724-4**, allowing the fluid delivery hub **724** to progress to further envelope the BFS vial neck **708** (e.g., the BFS vial neck **708** advancing further into the interior volume **724-3**). In some embodiments, the BFS vial neck **708** may comprise a plastic and/or polymer having a higher degree of compressibility than that of the fluid delivery hub **724**, such that the progression of the radial flanges **708-1**, **708-2** along the inside walls of the interior volume **724-3** may cause a radially inward compression and/or deformation of the flanges **708-1**, **708-2**. In some embodiments, such compression may be elastic such that upon encountering an increased interior diameter at the location of either of the channels **724-4**, **724-5** the flanges **708-1**, **708-2** may spring back radially outward to seat within the channels **724-4**, **724-5**. According to some embodiments, the seal **770** may be affixed and/or coupled to the administration member **760** such that axial force applied to the seal **770** causes the administration member **760** to move (e.g., as depicted in FIG. **7**A and FIG. **7**B).

[0060] In some embodiments, fewer or more components **708**, **708-1**, **708-2**, **712**, **712-1**, **724**, **724-3**, **724-4**, **724-5**, **726**, **728**, **728-1**, **744**, **744-3**, **746**, **748**, **748-1**, **760**, **760-1**, **760-2**, **760-3**, **770** and/or various configurations of the depicted components **708**, **708-1**, **708-2**, **712**, **712-1**, **724**, **724-3**, **724-4**, **724-5**, **726**, **728**, **728-1**, **744**, **744-3**, **746**, **748**, **748-1**, **760**, **760-1**, **760-2**, **760-3**, **770** may be included in the fluid delivery manifold system **700** without deviating from the scope of embodiments described herein. In some embodiments, the components **708**, **708-1**, **708-2**, **712**, **712-1**, **724**, **724-3**, **724-4**, **724-5**, **726**, **728**, **728-1**, **744**, **744-3**, **746**, **748**, **748-1**, **760**, **760-1**, **760-2**, **760-3**, **770** may be similar in configuration and/or functionality to similarly named and/or numbered components as described herein. In some embodiments, the fluid delivery manifold system **700** (and/or portion and/or component **708**, **708-1**, **708-2**, **712**, **712-1**, **724**, **724-3**, **724-4**, **724-5**, **726**, **728**, **728-1**, **744**, **744-3**, **746**, **748**, **748-1**, **760**, **760-1**, **760-2**, **760-3**, **770** thereof) may be utilized in accordance with the method **900** of FIG. **9** herein, and/or portions thereof.

[0061] Referring now to FIG. **8**A and FIG. **8**B, cross-sectional views of a fluid delivery manifold system **800** according to some embodiments are shown. In some embodiments, the fluid delivery manifold system **800** may be depicted in various states of activation in FIG. **8**A and FIG. **8**B. In FIG. **8**A, for example, the fluid delivery manifold system **800** may be depicted in a first or unengaged state while in FIG. **8**B the fluid delivery manifold system **800** may be depicted in a second or engaged state. According to some embodiments, a user (not shown) may progress the fluid delivery manifold system **800** from the first state to the second state by selective application of axial force that ultimately causes a fluid to be delivered to a recipient (not shown; e.g., a therapeutic and/or medical fluid to be delivered to a patient).

[0062] According to some embodiments, the fluid delivery manifold system **800** may comprise a BFS vial neck **808** having a first or distal exterior radial flange **808-1** and a second or proximate exterior radial flange **808-2**. In some embodiments, a fluid (not explicitly shown) may reside in and/or travel through a fluid passage **812** within the BFS vial neck **808** and/or may be contained by a fluid seal **812-1**. The fluid seal **812-1** may be integral to the BFS vial neck **808** such as by being formed during a BFS manufacturing process together with the BFS vial neck **808** or may comprise a separate element coupled to seal the BFS vial neck **808** and attendant fluid passage **812** thereof.

[0063] In some embodiments, the fluid delivery manifold system **800** may comprise a fluid delivery hub **824** coupled to the BFS vial neck **808**. The BFS vial neck **808** may, for example, be at least partially inserted into an interior volume **824-3** defined by the fluid delivery hub **824**, e.g., as shown in FIG. **8**A. According to some embodiments, the BFS vial neck **808** may have been inserted to a first position within the interior volume **824-3** as shown in FIG. **8**A, such that the first radial flange **808-1** engages with and/or becomes seated in a first or proximate radial channel **824-4** disposed on an inside surface of the interior volume **824-3**. As depicted in FIG. **8**B, continued advancement of the BFS vial neck **808** into the interior volume **824-3** may cause the first radial flange **808-1** to disengage with and/or move out of the first radial channel **824-4** and travel along the inside surface of the interior volume **824-3** until it becomes engaged with and/or seated in a second or distal radial channel **824-5** (as shown in FIG. **8**B). Upon achieving such a second and/or additional level of insertion into the interior volume **824-3**, in some embodiments the second radial flange **808-2** may become engaged with and/or seated in the first radial channel **824-4** (also as shown in FIG. **8**B). In such a manner, for example, at the first state shown in FIG. **8**A the BFS vial neck **708** may be secured to the fluid delivery hub **824** at a first position (e.g., via engagement of the first radial flange **808-1** with the first radial channel **824-4**; e.g., a transport, assembly, and/or pre-engagement position) while at the second state shown in FIG. **8**B the BFS vial neck **808** may be secured to the fluid delivery hub **824** at a second position (e.g., via engagement of the first radial flange **808-1** with the second radial channel **824-5** and engagement of the second radial flange **808-2** with the first radial channel **824-4**; e.g., an engagement position). In some embodiments, the interior volume **824-3** (and/or the channels **824-4**, **824-5** thereof) may be formed and/or defined by a hub body **826**. According to some embodiments, the hub body **826** may comprise and/or define a seat **828** disposed opposite to the interior volume **824-3** and/or a bore **828-1** extending between the seat **828** and the interior volume **824-3**.

[0064] In some embodiments, the fluid delivery manifold system **800** may comprise a safety cap **844** defining a lower void **844-3** into which the hub body **826** is at least partially inserted and/or disposed (e.g., as shown in FIG. **8**A) and/or an upper void **844-4**. The lower void **844-3** may comprise an inside diameter equivalent to (or slightly smaller than) an outside diameter of the hub body **826**, for example, such as permitting for a desired fit therebetween. According to some embodiments, the lower void **844-3** may be defined by a cap body **846** that may, for example, be comprise a lower body portion **846-1** having a first exterior diameter and an upper body portion **846-2** having a second exterior diameter. According to some embodiments, in the case that the second exterior diameter is smaller than the first exterior diameter, the cap body **846** may comprise and/or define (e.g., at a transition between the lower cap body **846-1** and the upper cap body **846-2**) an exterior flange **846-4**. In some embodiments, the upper void **844-4** may be defined within the upper body portion **846-2** and the lower void **844-3** may be defined within the lower body portion **846-1**.

[0065] According to some embodiments, the fluid delivery manifold system **800** may comprise an administration member **860** (such as a needle or tube) having a first or proximate end **860-1** and a second or distal end **860-2**. In some embodiments, the administration member **860** may be hollow and/or may otherwise define a fluid channel **860-3** extending from the first end **860-1** to the second end **860-3**. In some embodiments, the administration member **860** may pass through and/or be coupled to a seal **870** disposed in the lower void **844-3**. The seal **870** may comprise, for example, an annular rubber or thermoplastic element that may be deformable or pliable. According to some embodiments, either or both of the first end **860-1** and the second end **860-2** of the administration member **860** may comprise a tip, point, prong, blade, and/or other feature and/or configuration, such as to pierce the fluid seal **812-1** (e.g., in the case of the fires end **860-1**) or pierce an administration surface (such as skin; not shown; e.g., in the case of the second end **860-2**).

[0066] In some embodiments, and as depicted in FIG. **8**A, at the first or disengaged state of the fluid delivery manifold system **800** the first end **860-1** of the administration member **860** may be disposed in the interior volume **824-3** of the fluid delivery hub **824** and the second end **860-2** of the administration member **860** may be disposed in the upper void **848** of the safety cap **844**. The administration member **860** may pass through, for example, each of the bore **828-1** (and the seat **828**), the seal **870**, and the passage **848-1**. According to some embodiments, axial force may be applied (e.g., to the exterior flange **846-4**) to transition the fluid delivery manifold system **800** to the second or engaged state of FIG. **8**B where the first end **860-1** of the administration member **860** has pierced the fluid seal **812-1** and is disposed within the fluid passage **812** of the BFS vial neck **808** and where the seal **870** is seated in, engaged with, and/or coupled to the seat **828**. In such a manner, for example, the fluid delivery manifold system **800** may be engaged to provide the fluid from the fluid passage **812**, through the fluid channel **860-3**, and into a recipient (not shown). In some embodiments, the safety cap **844** may be removed to expose the administration member **860** and/or the second end **860-2** thereof for administration of the fluid. According to some embodiments, axial force (downward, as-oriented in FIG. **8**A and FIG. **8**B) may cause the seal **870** to deform and/or compress into the seat **828**. Axial force applied to the safety cap **844** (e.g., to the exterior flange **846-4** thereof) may cause the cap body **846** (and/or the ceiling of the interior volume **824-3**) to act upon the seal **870** and urge the seal **870** axially downward (e.g., as depicted transitioning from FIG. **8**A to

FIG. **8**B). In some embodiments, once the seal **870** contacts the seat **828** it may be forced therein and/or may be deformed or compressed into the seat **828**.

[0067] According to some embodiments, upon the seal **870** achieving a degree of compression and/or deformation, force may be transferred to the hub body **826**, causing the first radial flange **808-1** of the BFS vial neck **808** to unseat and/or disengage from the first radial channel **824-4**, allowing the fluid delivery hub **824** to progress to further envelope the BFS vial neck **808** (e.g., the BFS vial neck **808** advancing further into the interior volume **824-3**). In some embodiments, the BFS vial neck **808** may comprise a plastic and/or polymer having a higher degree of compressibility than that of the fluid delivery hub **824**, such that the progression of the radial flanges **808-1**, **808-2** along the inside walls of the interior volume **824-3** may cause a radially inward compression and/or deformation of the flanges **808-1**, **808-2**. In some embodiments, such compression may be elastic such that upon encountering an increased interior diameter at the location of either of the channels **824-4**, **824-5** the flanges **808-1**, **808-2** may spring back radially outward to seat within the channels **824-4**, **824-5**. According to some embodiments, the seal **870** may be affixed and/or coupled to the administration member **860** such that axial force applied to the seal **870** causes the administration member **860** to move (e.g., as depicted in FIG. **8**A and FIG. **8**B).

[0068] In some embodiments, fewer or more components **808**, **808-1**, **808-2**, **812**, **812-1**, **824**, **824-3**, **824-4**, **824-5**, **826**, **828**, **828-1**, **844**, **844-3**, **844-4**, **846**, **846-1**, **846-2**, **846-4**, **860**, **860-1**, **860-2**, **860-3**, **870** and/or various configurations of the depicted components **808**, **808-1**, **808-2**, **812**, **812-1**, **824**, **824-3**, **824-4**, **824-5**, **826**, **828**, **828-1**, **844**, **844-3**, **844-4**, **846**, **846-1**, **846-2**, **846-4**, **860**, **860-1**, **860-2**, **860-3**, **870** may be included in the fluid delivery manifold system **800** without deviating from the scope of embodiments described herein. In some embodiments, the components **808**, **808-1**, **808-2**, **812**, **812-1**, **824**, **824-3**, **824-4**, **824-5**, **826**, **828**, **828-1**, **844**, **844-3**, **844-4**, **846**, **846-1**, **846-2**, **846-4**, **860**, **860-1**, **860-2**, **860-3**, **870** may be similar in configuration and/or functionality to similarly named and/or numbered components as described herein. In some embodiments, the fluid delivery manifold system **800** (and/or portion and/or component **808**, **808-1**, **808-2**, **812**, **812-1**, **824**, **824-3**, **824-4**, **824-5**, **826**, **828**, **828-1**, **844**, **844-3**, **844-4**, **846**, **846-1**, **846-2**, **846-4**, **860**, **860-1**, **860-2**, **860-3**, **870** thereof) may be utilized in accordance with the method **900** of FIG. **9** herein, and/or portions thereof.

III. Fluid Delivery Manifold Methods

[0069] FIG. **9** is a flow diagram of a method **900** according to some embodiments. The method **900** may, for example, illustrate an exemplary use of the various fluid delivery manifold systems and/or components thereof, as described herein. In some embodiments, the method **900** may comprise multiple related and/or consecutive processes such as actions associated with a first process "A" and a second process "B". According to some embodiments, the different processes "A", "B" may be performed by different entities and/or at different times and/or locations. The process diagrams and flow diagrams described herein do not necessarily imply a fixed order to any depicted actions, steps, and/or procedures, and embodiments may generally be performed in any order that is practicable unless otherwise and specifically noted. While the order of actions, steps, and/or procedures described herein is generally not fixed, in some embodiments, actions, steps, and/or procedures may be specifically performed in the order listed, depicted, and/or described and/or may be performed in response to any previously listed, depicted, and/or described action, step, and/or procedure.

[0070] According to some embodiments, the first process "A" may comprise a manufacturing and/or assembly process. In some embodiments, the first process "A" and/or the method **900** may comprise assembling a BFS vial package with a fluid delivery manifold, at **902**. A plurality of BFS vials connected in a package may be mated with corresponding fluid delivery hub members that are also interconnected, for example. According to some embodiments, the assembling may include an aligning of the respective BFS vials (or necks thereof) with individual fluid delivery hub components and applying a first axial force to cause mating of the BFS vials and fluid delivery hubs.

[0071] In some embodiments, the method **900** may comprise coupling a plurality of administration members (e.g., needles, droppers, nozzles) and respective fluid seals to the fluid delivery manifold, at **904**. Each fluid delivery hub may comprise a bore in which an administration member is inserted, for example. According to some embodiments, the method **900** may comprise assembling a safety cap manifold with the fluid delivery manifold, at **906**. In some embodiments, the assembling may include an aligning of the respective fluid delivery hubs with individual safety caps and applying a second axial force to cause mating of the fluid delivery hubs and the safety caps. According to some embodiments, a plurality of interconnected safety caps may be aligned, for example, and engaged with the fluid delivery hubs of the fluid delivery manifold. According to some embodiments, the coupling and/or mating of the safety caps with the fluid delivery hubs may cause the administration members and/or fluid seals to become encapsulated, retained, and/or restricted. In some embodiments, each administration member and each fluid seal may be housed within a respective safety cap and/or between the safety cap and a respective fluid delivery hub. In some embodiments, the assembly and coupling actions may occur in an aseptic environment, such as part of an aseptic BFS manufacturing process that also forms, fills, and seals the BFS vials.

[0072] According to some embodiments, the second process "B" may comprise an administration process. In some embodiments, the second process "B" and/or the method **900** may comprise activating a unit of the fluid delivery manifold system, at **908**. A third axial force may be applied to axially compress a unit comprising a mated pairing of a BFS vial, a fluid delivery hub, an administration member, a seal, and a safety cap, for example, thereby causing an activation thereof. The third axial force may cause the administration member to pierce a seal of the BFS vial, for example, and/or may cause the fluid seal to become seated and/or coupled to the fluid delivery hub. According to some embodiments, the activating may comprise removing the unit from the respective package and manifolds, such as by application of a twisting, shear, and/or rotational force thereto (e.g., with respect to the other units of the assembled fluid delivery manifold system).

[0073] In some embodiments, the method **900** may comprise removing the safety cap of the unit, at **910**. Axial separation force may be applied to separate the safety cap from the fluid delivery hub of the unit, for example, exposing the administration member such as a needle, nozzle, dropper, or the like. According to some embodiments, the method **900** may comprise administering a dose of fluid, at **912**. The administration element of the unit may be engaged with a patient, for example, and a collapsible and/or integral reservoir of the BFS vial may be squeezed (e.g., via application of inward radial force) to force fluid therefrom. According to some embodiments, the fluid may be forced in an antegrade axial direction such that it displaces a valve flap of a one-way valve, thereby allowing the fluid to proceed axially into the administration element and be delivered to the patient. In some embodiments, the method **900** may comprise replacing the safety cap, at **914**. According to some embodiments, the method **900** may comprise disposing of the unit, at **916**. In such a manner, for example, a low-cost, easily transported and more easily manufactured, assembled, and stored fluid delivery system may be provided that allows unskilled users to administer and/or self-administer various fluids such as medical and/or therapeutic substances.

IV. Rules of Interpretation

[0074] Throughout the description herein and unless otherwise specified, the following terms may include and/or encompass the example meanings provided. These terms and illustrative example meanings are provided to clarify the language selected to describe embodiments both in the specification and in the appended claims, and accordingly, are not intended to be generally limiting. While not generally limiting and while not limiting for all described embodiments, in some embodiments, the terms are specifically limited to the example definitions and/or examples provided. Other terms are defined throughout the present description.

[0075] Numerous embodiments are described in this patent application, and are presented for illustrative purposes only. The described embodiments are not, and are not intended to be, limiting in any sense. The presently disclosed invention(s) are widely applicable to numerous embodiments, as is readily apparent from the disclosure. One of ordinary skill in the art will recognize that the disclosed invention(s) may be practiced with various modifications and alterations, such as structural, logical, software, and electrical modifications. Although particular features of the disclosed invention(s) may be described with reference to one or more particular embodiments and/or drawings, it should be understood that such features are not limited to usage in the one or more particular embodiments or drawings with reference to which they are described, unless expressly specified otherwise.

[0076] Devices that are in communication with each other need not be in continuous communication with each other, unless expressly specified otherwise.

[0077] A description of an embodiment with several components or features does not imply that all or even any of such components and/or features are required. On the contrary, a variety of optional components are described to illustrate the wide variety of possible embodiments of the present invention(s). Unless otherwise specified explicitly, no component and/or feature is essential or required.

[0078] Further, although process steps, algorithms or the like may be described in a sequential order, such processes may be configured to work in different orders. In other words, any sequence or order of steps that may be explicitly described does not necessarily indicate a requirement that the steps be performed in that order. The steps of processes described herein may be performed in any order practical. Further, some steps may be performed simultaneously despite being described or implied as occurring non-simultaneously (e.g., because one step is described after the other step). Moreover, the illustration of a process by its depiction in a drawing does not imply that the illustrated process is exclusive of other variations and modifications thereto, does not imply that the illustrated process or any of its steps are necessary to the invention, and does not imply that the illustrated process is preferred.

[0079] The present disclosure provides, to one of ordinary skill in the art, an enabling description of several embodiments and/or inventions. Some of these embodiments and/or inventions may not be claimed in the present application, but may nevertheless be claimed in one or more continuing applications that claim the benefit of priority of the present application. Applicants intend to file additional applications to pursue patents for subject matter that has been disclosed and enabled but not claimed in the present application.

[0080] It will be understood that various modifications can be made to the embodiments of the present disclosure herein without departing from the scope thereof. Therefore, the above description should not be construed as limiting the disclosure, but merely as embodiments thereof. Those skilled in the art will envision other modifications within the scope of the invention as defined by the claims appended hereto.

[0081] While several embodiments of the present disclosure have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the functions and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the present disclosure. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the teachings of the present disclosure is/are used.

[0082] Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the disclosure described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, the disclosure may be practiced otherwise than as specifically described and claimed. The present disclosure is directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the scope of the present disclosure.

[0083] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

[0084] The indefinite articles "a" and "an," as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean "at least one."

[0085] The phrase "and/or," as used herein in the specification and in the claims, should be understood to mean "either or both" of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Other elements may optionally be present other than the elements specifically identified by the "and/or" clause, whether related or unrelated to those elements specifically identified, unless clearly indicated to the contrary.

[0086] Reference throughout this specification to "one embodiment" or "an embodiment" means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases "in one embodiment" or "in an embodiment" in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

[0087] The terms and expressions which have been employed herein are used as terms of description and not of limitation, and there is no intention, in the use of such terms and expressions, of excluding any equivalents of the features shown and described (or portions thereof), and it is recognized that various modifications are possible within the scope of the claims. Accordingly, the claims are intended to cover all such equivalents.

[0088] Various modifications of the invention and many further embodiments thereof, in addition to those shown and described herein, will become apparent to those skilled in the art from the full contents of this document, including references to the scientific and patent literature cited herein. The subject matter herein contains important information, exemplification and guidance that can be adapted to the practice of this invention in its various embodiments and equivalents thereof.

What is claimed is:

1. A fluid delivery system for delivery of a fluid agent, the fluid delivery system comprising:

an individual BFS vial containing a single dose of a fluid agent, the BFS vial comprising a radially compressible reservoir containing the single dose of the fluid agent and comprising a cylindrical neck defining a fluid channel in communication with the radially compressible reservoir and the BFS vial comprising at least one mating feature disposed on an exterior surface thereof;

a fluid delivery hub defining an interior volume having one or more mating features disposed on an interior surface therein, and each of the one or more mating features being coupled to a corresponding one of the at least one mating feature of the exterior surface of the BFS vial; and

an administration member being coupled to the fluid delivery hub.

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